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Solution in Lorentz gauge

Real space

$$\mathbf{A}_{\sigma,\ell}^L(\mathbf{r}) = f_{|\ell|}(\rho) e^{i\ell\phi} e^{iq_0 z} \hat{\mathbf{e}}_{\sigma}$$

$$f_{\ell}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2}{\pi|\ell|!}} L_{p=0}^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

Angular spectrum

$$\mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_{\parallel}) = \mathcal{A}_{\ell}^L(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{\sigma} = \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{i\ell\phi_{q_{\parallel}}} \hat{\mathbf{e}}_{\sigma}$$

$$\tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) = (-i)^{|\ell|} \frac{A_0^{\text{LG}} W_0^2 \pi}{\Omega \sqrt{\pi|\ell|!}/2} \left(\frac{q_{\parallel} W_0}{\sqrt{2}} \right)^{|\ell|} e^{-\frac{q_{\parallel}^2 W_0^2}{4}}$$

Vector potential under
Lorentz gauge in ...

Expand $f_{|\ell|}(\rho) e^{i\ell\phi}$ in
plane wave basis

$$\int_0^{2\pi} d\phi \int_0^{\infty} d\rho \rho f_{\ell}(\rho) e^{i\ell\phi} e^{-i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} = \tilde{K}_{\ell}(\mathbf{q}_{\parallel}) e^{i\ell\phi_{q_{\parallel}}} \left\{ \begin{array}{ll} \ell \geq 0 & (-i)^{+\ell} \frac{A_0^{\text{LG}} W_0^2 \pi}{\Omega \sqrt{\pi|\ell|!}/2} \left(\frac{1}{\sqrt{2}} q_{\parallel} W_0 \right)^{+\ell} e^{-\frac{1}{4} q_{\parallel}^2 W_0^2} e^{i\ell\phi_{q_{\parallel}}} \hat{\mathbf{e}}_{\sigma} \\ \ell < 0 & (-i)^{-\ell} \frac{A_0^{\text{LG}} W_0^2 \pi}{\Omega \sqrt{\pi|\ell|!}/2} \left(\frac{1}{\sqrt{2}} q_{\parallel} W_0 \right)^{-\ell} e^{-\frac{1}{4} q_{\parallel}^2 W_0^2} e^{i\ell\phi_{q_{\parallel}}} \hat{\mathbf{e}}_{\sigma} \end{array} \right\} = \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel})$$

Only phase $e^{i\ell\phi_{q_{\parallel}}}$ carrying OAM

First appearance of SOI (appearance of σ in phase)

under Coulomb gauge ... Real space

$$\mathbf{A}_{\sigma,\ell}(r) \equiv \mathbf{A}_{\sigma,\ell}^C(r) \approx \tilde{A}_{\ell}^{\parallel}(r) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell}^Z(r) \hat{\mathbf{z}}$$

$$\tilde{A}_{\ell}^{\parallel}(r) \approx f_{\ell}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$\tilde{A}_{\sigma,\ell}^Z(r) \approx \frac{i}{\sqrt{2}q_0\rho} \left[(|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right] f_{\ell}(\rho) e^{iJ\phi} e^{iq_0 z}$$

Angular spectrum

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) \equiv \mathcal{A}_{\sigma,\ell}^C(r) \approx \tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell}^Z(\mathbf{q}_{\parallel}) \hat{\mathbf{z}}$$

$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}} \hat{\mathbf{e}}_{\sigma}$$

$$\tilde{\mathcal{A}}_{\sigma,\ell}^Z(\mathbf{q}_{\parallel}) \approx \frac{-\sin\theta_q}{\sqrt{2}} \tilde{F}_{|\ell|}(q_{\parallel}) e^{iJ\phi_{q_{\parallel}}} \hat{\mathbf{z}}$$

SOI term, $J \equiv \sigma + \ell$, appear when gauge transforming

$$\mathbf{A}_{\sigma,\ell}(r) = \mathbf{A}_{\sigma,\ell}^L(r) + \frac{\nabla(\nabla \cdot \mathbf{A}_{\sigma,\ell}^L(r))}{q_0^2}$$

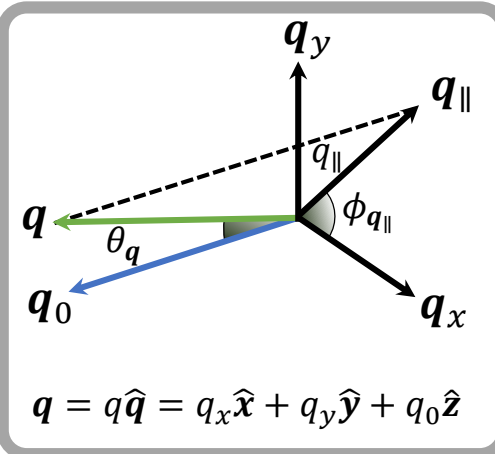
$$= \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z} + \frac{1}{q_0^2} \nabla \left(\nabla \cdot \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z} \right)$$

$$= \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z} + \frac{1}{q_0^2} \nabla \left(\left(\frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}} + \frac{\partial}{\partial z} \hat{\mathbf{z}} \right) \cdot \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_{\parallel}) e^{i(q_x x + q_y y + q_0 z)} \right)$$

$$= \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z} + \frac{1}{q_0^2} \nabla \left(\frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} (iq_x \hat{\mathbf{x}} + iq_y \hat{\mathbf{y}}) \cdot \hat{\mathbf{e}}_{\sigma} \mathcal{A}_{\ell}^L(\mathbf{q}_{\parallel}) e^{i(q_x x + q_y y + q_0 z)} \right)$$



$$iq_{\parallel} (\cos\phi_{q_{\parallel}} \hat{\mathbf{x}} + \sin\phi_{q_{\parallel}} \hat{\mathbf{y}}) \cdot \hat{\mathbf{e}}_{\sigma} = \frac{i}{\sqrt{2}} q_{\parallel} (\cos\phi_{q_{\parallel}} + i\sigma \sin\phi_{q_{\parallel}}) \cdot \hat{\mathbf{e}}_{\sigma} = \frac{i}{\sqrt{2}} q \sin\theta_q e^{i\sigma\phi_{q_{\parallel}}}$$



First appearance of SOI (in longitudinal component)

under Coulomb gauge ... Real space

$$\mathbf{A}_{\sigma,\ell}(r) \equiv \mathbf{A}_{\sigma,\ell}^C(r) \approx \tilde{A}_{\ell}^{\parallel}(r) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell}^Z(r) \hat{\mathbf{z}}$$

$$\tilde{A}_{\ell}^{\parallel}(r) \approx f_{\ell}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$\tilde{A}_{\sigma,\ell}^Z(r) \approx \frac{i}{\sqrt{2}q_0\rho} \left[(|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right] f_{\ell}(\rho) e^{iJ\phi} e^{iq_0 z}$$

Angular spectrum

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) \equiv \mathcal{A}_{\sigma,\ell}^C(r) \approx \tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell}^Z(\mathbf{q}_{\parallel}) \hat{\mathbf{z}}$$

$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}} \hat{\mathbf{e}}_{\sigma}$$

$$\tilde{\mathcal{A}}_{\sigma,\ell}^Z(\mathbf{q}_{\parallel}) \approx \frac{-\sin\theta_q}{\sqrt{2}} \tilde{F}_{|\ell|}(q_{\parallel}) e^{iJ\phi_{q_{\parallel}}} \hat{\mathbf{z}}$$

J determine the $\phi_{q_{\parallel}}$ dependent phase in longitudinal component

$$\mathbf{A}_{\sigma,\ell}(r) = \mathbf{A}_{\sigma,\ell}^L(r) + \frac{\nabla(\nabla \cdot \mathbf{A}_{\sigma,\ell}^L(r))}{q_0^2}$$

$$= \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z} + \frac{1}{q_0^2} \nabla \left(\frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \left[\frac{i}{\sqrt{2}} q \sin\theta_q e^{i\sigma\phi_{q_{\parallel}}} \right] \mathcal{A}_{\ell}^L(\mathbf{q}_{\parallel}) e^{i(q_x x + q_y y + q_0 z)} \right)$$

$$= \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z} + \frac{1}{q_0^2} \left(\frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}} + \frac{\partial}{\partial z} \hat{\mathbf{z}} \right) \left(\frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \left[\frac{i}{\sqrt{2}} q \sin\theta_q e^{i\sigma\phi_{q_{\parallel}}} \right] \mathcal{A}_{\ell}^L(\mathbf{q}_{\parallel}) e^{i(q_x x + q_y y + q_0 z)} \right)$$

$$= \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z} + \frac{1}{q_0^2 (2\pi)^2} \int d\mathbf{q}_{\parallel} i q \hat{\mathbf{q}} \left[\frac{i}{\sqrt{2}} q \sin\theta_q e^{i\sigma\phi_{q_{\parallel}}} \right] \mathcal{A}_{\ell}^L(\mathbf{q}_{\parallel}) e^{i(q_x x + q_y y + q_0 z)}$$

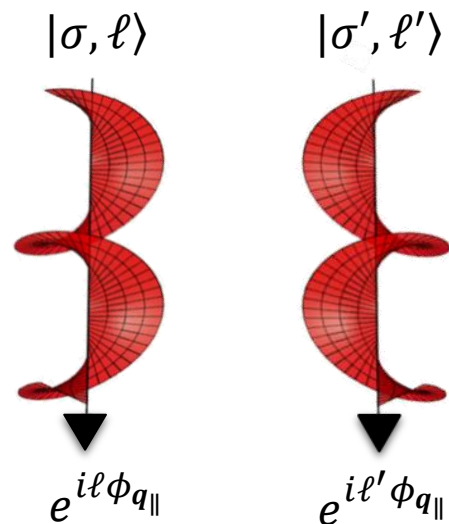
$$= \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \left\{ \hat{\mathbf{e}}_{\sigma} - \frac{q^2}{q_0^2} \left[\frac{1}{\sqrt{2}} \sin\theta_q e^{i\sigma\phi_{q_{\parallel}}} \right] \hat{\mathbf{q}} \right\} \mathcal{A}_{\ell}^L(\mathbf{q}_{\parallel}) e^{i(q_x x + q_y y + q_0 z)} = \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z}$$

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) = \left\{ \hat{\mathbf{e}}_{\sigma} - \frac{q^2}{q_0^2} \left[\frac{1}{\sqrt{2}} \sin\theta_q e^{i\sigma\phi_{q_{\parallel}}} \right] \hat{\mathbf{q}} \right\} \mathcal{A}_{\ell}^L(\mathbf{q}_{\parallel}) = \left[\hat{\mathbf{e}}_{\sigma} - \left(\frac{q}{q_0} \right)^2 (\hat{\mathbf{e}}_{\sigma} \cdot \hat{\mathbf{q}}_{\parallel}) \hat{\mathbf{q}} \right] \mathcal{A}_{\ell}^L(\mathbf{q}_{\parallel})$$

$$\text{where } \mathcal{A}_{\ell}^L(\mathbf{q}_{\parallel}) = \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}}$$

Exciton Transition with Beating of Vector Vortex Beams

1. Analogy signal additive mixing to vector vortex beams (VVB)



Vector vortex beams :

$$\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) = \cos\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma',\ell'}(\mathbf{q}_{\parallel})$$

Single twisted light : $\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) = \tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) + \tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel})$

$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{i\ell\phi_{q_{\parallel}}} \hat{\mathbf{e}}_{\sigma}$$

$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel}) \approx \frac{-\sin\theta_q}{\sqrt{2}} \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{iJ\phi_{q_{\parallel}}} \hat{\mathbf{z}} \quad J \equiv \sigma + \ell$$

Amplitude :

$$\cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) \longrightarrow \tilde{A}$$

$$\sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(\mathbf{q}_{\parallel}) \longrightarrow \tilde{A}'$$

Phase :

$$e^{i\ell\phi_{q_{\parallel}}} \longrightarrow e^{ik\phi}$$

$$e^{iJ\phi_{q_{\parallel}}}$$

Signal additive mixing :

$$\text{Signal 1 } (\phi, t) = \text{Re}[\tilde{A} e^{i(k\phi - \omega t)}]$$

$$\text{Signal 2 } (\phi, t) = \text{Re}[\tilde{A}' e^{i(k'\phi - \omega t + \alpha)}]$$

$$\text{Sum} = \text{Re}[\tilde{A} e^{i(k\phi - \omega t)} + \tilde{A}' e^{i(k'\phi - \omega t + \alpha)}]$$

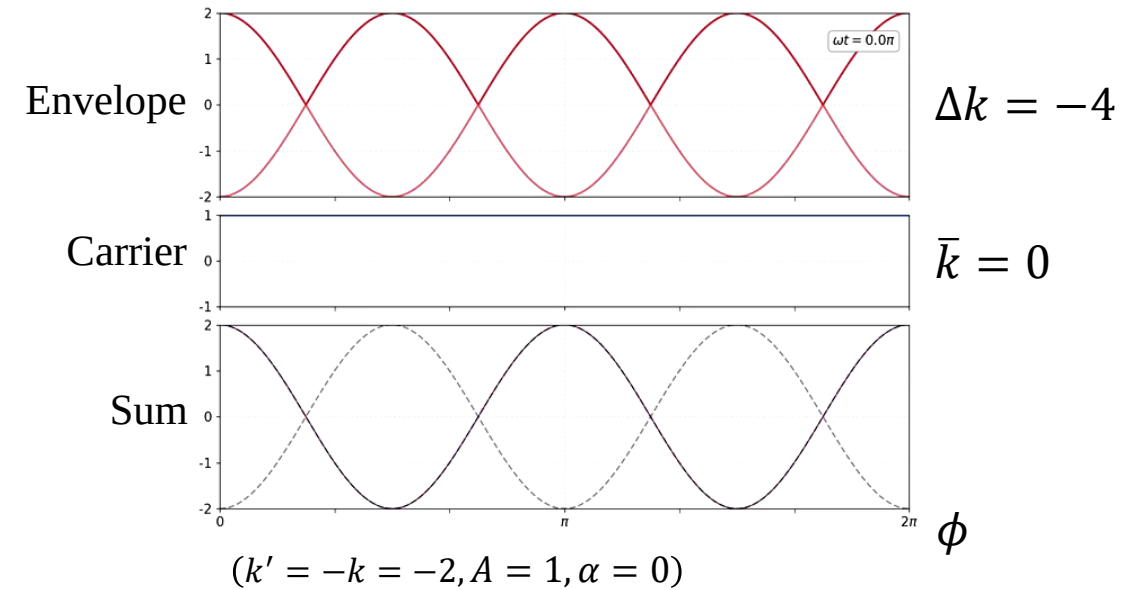
1. Signal additive mixing (beating/rotating)

1. Standing wave : $\tilde{A} = \tilde{A}' = A$ and $k' = -k$

$$\text{Sum} = \text{Re} \left[\underbrace{2A \cos\left(\frac{\Delta k}{2} \phi + \frac{\alpha}{2}\right)}_{\text{Envelope}} \underbrace{e^{i(-\omega t + \frac{\alpha}{2})}}_{\text{Carrier}} \right]$$

$(\Delta k \equiv k' - k)$

➡ Number of nodes = beating wavenumber $|\Delta k|$

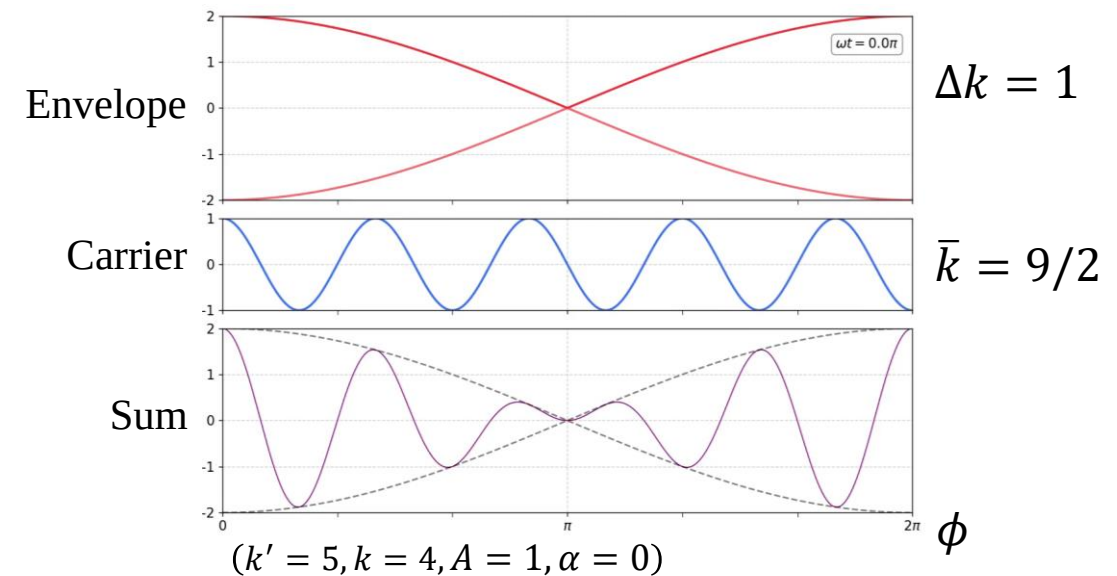


2. Signal beating : $\tilde{A} = \tilde{A}' = A$ and $k' \neq -k$

$$\text{Sum} = \text{Re} \left[\underbrace{2A \cos\left(\frac{\Delta k}{2} \phi + \frac{\alpha}{2}\right)}_{\text{Envelope}} \underbrace{e^{i(\bar{k} \phi - \omega t + \frac{\alpha}{2})}}_{\text{Carrier}} \right]$$

$(\bar{k} \equiv \frac{k' + k}{2})$

➡ Non-zero phase velocity ($\bar{k} \neq 0$), rotating field

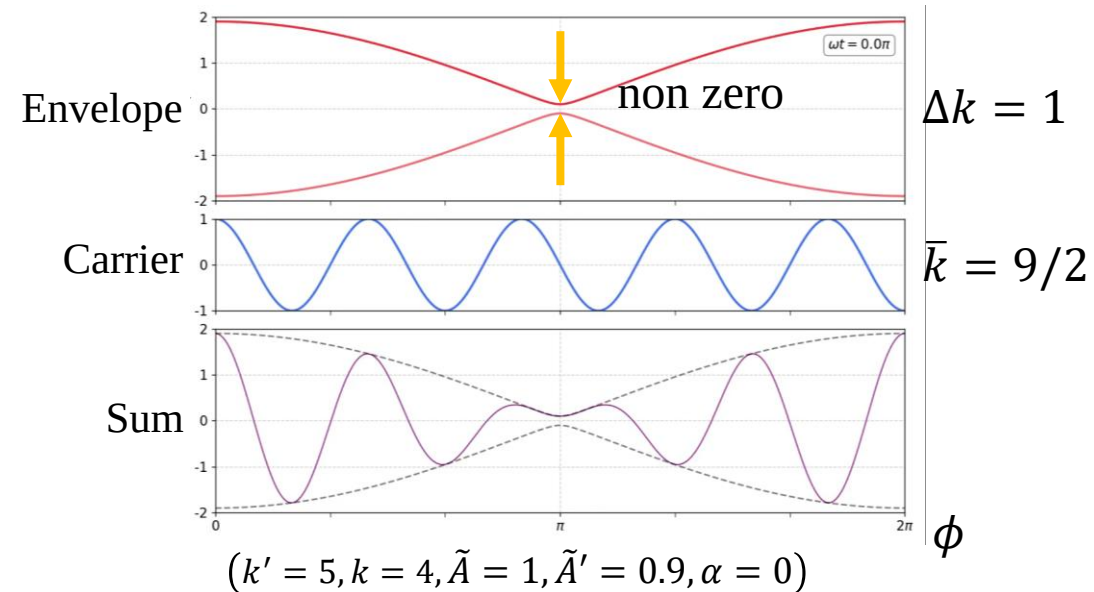


1. Signal additive mixing (amplitude mismatch with $|\tilde{A}| + |\tilde{A}'| \gg |\tilde{A}| - |\tilde{A}'|$)⁸

3. $\tilde{A} \neq \tilde{A}' \in \mathbb{R}$ and $k' \neq -k$

$$\begin{aligned} \text{Sum} &= \text{Re} \left\{ \left[(\tilde{A} + \tilde{A}') \cos\left(\frac{\Delta k}{2} \phi + \frac{\alpha}{2}\right) - i(\tilde{A} - \tilde{A}') \sin\left(\frac{\Delta k}{2} \phi + \frac{\alpha}{2}\right) \right] e^{i(\bar{k}\phi - \omega t + \frac{\alpha}{2})} \right\} \\ &\approx \text{Re} \left\{ \underbrace{(\tilde{A} + \tilde{A}') \cos\left(\frac{\Delta k}{2} \phi + \frac{\alpha}{2}\right)}_{\text{Envelope}} \underbrace{e^{i(\bar{k}\phi - \omega t + \frac{\alpha}{2})}}_{\text{Carrier}} \right\} \end{aligned}$$

➡ Within the approximation, the structure is similar to 2.

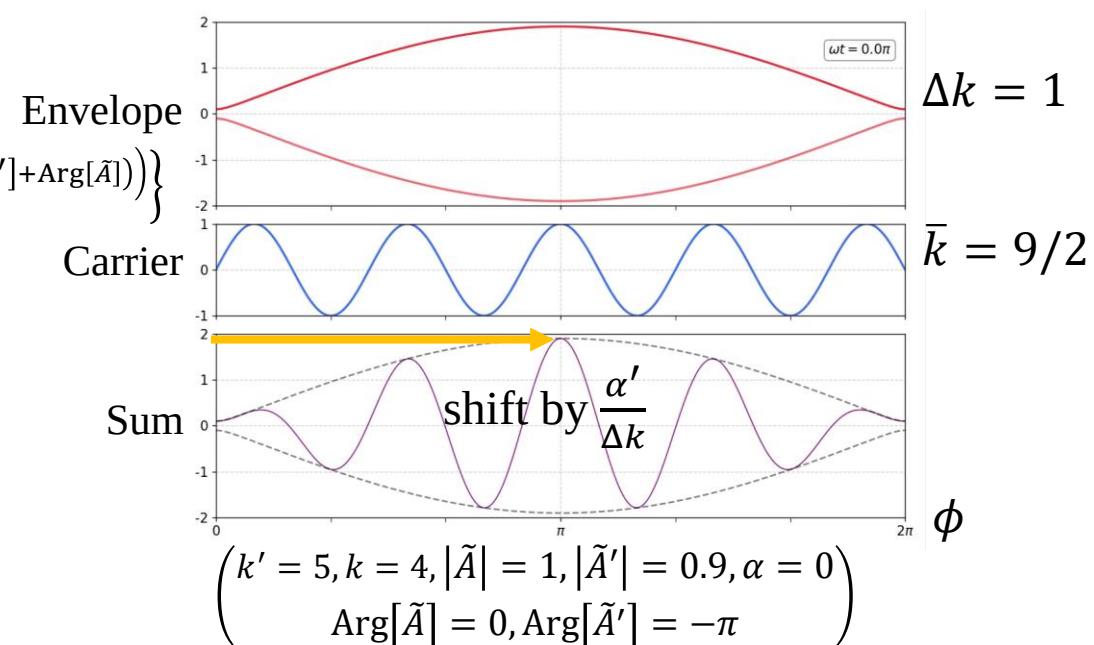


4. $\tilde{A} \neq \tilde{A}' \in \mathbb{C}$ and $k' \neq -k$

$$\begin{aligned} \text{Sum} &= \text{Re} \left\{ \left[(|\tilde{A}| + |\tilde{A}'|) \cos\left(\frac{\Delta k}{2} \phi + \frac{\alpha}{2}\right) - i(|\tilde{A}| - |\tilde{A}'|) \sin\left(\frac{\Delta k}{2} \phi + \frac{\alpha}{2}\right) \right] e^{i(\bar{k}\phi - \omega t + \frac{\alpha}{2} + \frac{1}{2}(\text{Arg}[\tilde{A}'] + \text{Arg}[\tilde{A}]))} \right\} \\ &\approx \text{Re} \left\{ \underbrace{(|\tilde{A}| + |\tilde{A}'|) \cos\left(\frac{\Delta k}{2} \phi + \frac{\alpha'}{2}\right)}_{\text{Envelope}} \underbrace{e^{i(\bar{k}\phi - \omega t + \frac{\alpha}{2} + \frac{1}{2}(\text{Arg}[\tilde{A}'] + \text{Arg}[\tilde{A}]))}}_{\text{Carrier}} \right\} \end{aligned}$$

$(\alpha' \equiv \alpha + \text{Arg}[\tilde{A}'] - \text{Arg}[\tilde{A}])$

➡ Relative phase α' change the intensity/phase



Longitudinal component of VVB

2. Analogy to beating formulation (with approximation)

Beating formulation :

$$\begin{aligned} \text{Sum} &= \text{Re}[\tilde{A}e^{i(k\phi - \omega t)} + \tilde{A}'e^{i(k'\phi - \omega t + \alpha)}] \\ &\approx \underbrace{\text{Re}\left\{(|\tilde{A}| + |\tilde{A}'|) \cos\left(\frac{\Delta k}{2}\phi + \frac{\alpha'}{2}\right)\right\}}_{\text{Envelope}} \underbrace{e^{i\left(\bar{k}\phi - \omega t + \frac{\alpha}{2} + \frac{1}{2}(\text{Arg}[\tilde{A}'] + \text{Arg}[\tilde{A}])\right)}}_{\text{Carrier}} \end{aligned}$$

Longitudinal component of VVB :

$$\begin{aligned} \tilde{\mathcal{A}}_{\sigma, \ell, \sigma', \ell'}^z(\mathbf{q}_{\parallel}, t; \alpha, \beta) &= \left[\cos\left(\frac{\beta}{2}\right) \frac{-\sin \theta_q}{\sqrt{2}} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i(J\phi_{q_{\parallel}} - \omega t)} + \sin\left(\frac{\beta}{2}\right) \frac{-\sin \theta_q}{\sqrt{2}} \tilde{F}_{|\ell'|}(q_{\parallel}) e^{i(J'\phi_{q_{\parallel}} - \omega t + \alpha)} \right] \hat{\mathbf{z}} \\ &\approx \underbrace{G_{|\ell|, |\ell'|}^+(q_{\parallel}; \beta) \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2}\right)}_{\text{Envelope}} \times \underbrace{e^{i\left(\bar{J}\phi_{q_{\parallel}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|)\right)}}_{\text{Carrier}} \hat{\mathbf{z}} \end{aligned}$$

node structure
adjust intensity
rotating field

Definition :

$$\begin{aligned} |\tilde{A}| \pm |\tilde{A}'| &\longrightarrow G_{|\ell|, |\ell'|}^{\pm}(q_{\parallel}; \beta) \equiv \left| -\frac{\sin \theta_q}{\sqrt{2}} \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(q_{\parallel}) \right| \pm \left| -\frac{\sin \theta_q}{\sqrt{2}} \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(q_{\parallel}) \right| \\ \Delta k \equiv k' - k \text{ and } \bar{k} \equiv \frac{k' + k}{2} &\longrightarrow J \equiv J' - J \text{ and } \bar{J} \equiv \frac{J' + J}{2} \\ \alpha' \equiv \alpha + \text{Arg}[\tilde{A}'] - \text{Arg}[\tilde{A}] &\longrightarrow \alpha_{|\ell|, |\ell'|} \equiv \alpha + \text{Arg}[\tilde{F}_{|\ell'|}(q_{\parallel})] - \text{Arg}[\tilde{F}_{|\ell|}(q_{\parallel})] = \alpha - \frac{\pi}{2}(|\ell'| - |\ell|) \end{aligned}$$

2. Analogy to beating formulation (without approximation)

Beating formulation :

Without assuming $|\tilde{A}| + |\tilde{A}'| \gg |\tilde{A}| - |\tilde{A}'|$

$$\text{Sum} = \text{Re} \left\{ \left[(|\tilde{A}| + |\tilde{A}'|) \cos \left(\frac{\Delta k}{2} \phi + \frac{\alpha}{2} \right) - \underbrace{i(|\tilde{A}| - |\tilde{A}'|) \sin \left(\frac{\Delta k}{2} \phi + \frac{\alpha}{2} \right)}_{\text{Previously neglected}} \right] e^{i \left(\bar{k} \phi - \omega t + \frac{\alpha}{2} + \frac{1}{2} (\text{Arg}[\tilde{A}'] + \text{Arg}[\tilde{A}]) \right)} \right\}$$

Longitudinal
component of VVB :

Without assuming $G_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \gg G_{|\ell|,|\ell'|}^-(q_{\parallel}; \beta)$

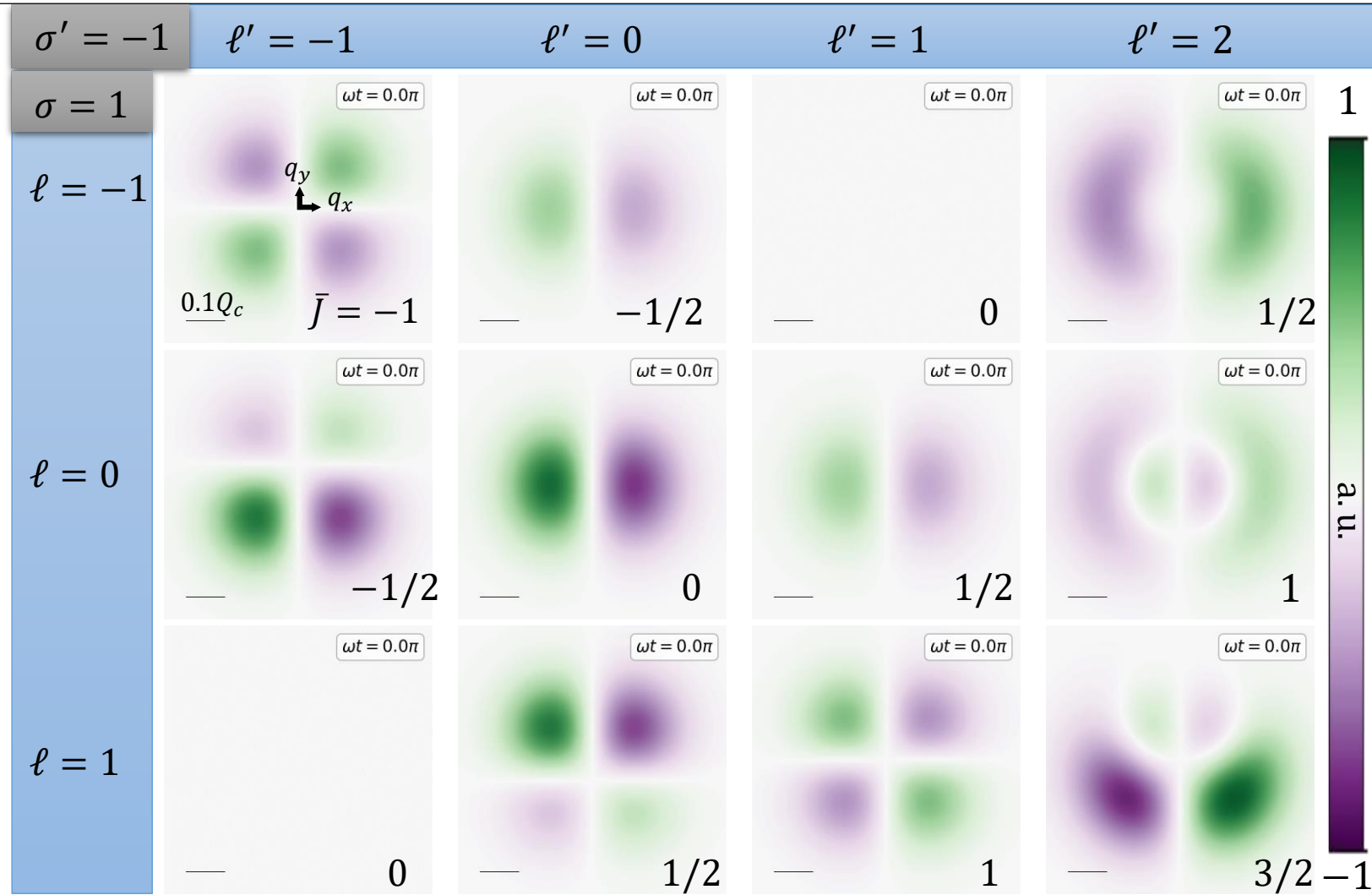
$$\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{q}_{\parallel}, t; \alpha, \beta) = \left[G_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \cos \left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2} \right) - \underbrace{i G_{|\ell|,|\ell'|}^-(q_{\parallel}; \beta) \sin \left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2} \right)}_{\text{Previously neglected}} \right] \times e^{i \left(\bar{J} \phi_{q_{\parallel}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4} (|\ell'| + |\ell|) \right)} \hat{\mathbf{z}}$$

2. Longitudinal component (rotating/standing)

$$\begin{aligned} \text{Re}[\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{q}_{\parallel}, t; \alpha, \beta)] &= \text{Re} \left\{ \left[G_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - i G_{|\ell|,|\ell'|}^-(q_{\parallel}; \beta) \sin\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \times e^{i\left(\bar{J}\phi_{q_{\parallel}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|)\right)} \right\} \hat{\mathbf{z}} \\ &\approx \underbrace{\text{Re} \left[G_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right]}_{\text{Envelope}} \underbrace{\times e^{i\left(\bar{J}\phi_{q_{\parallel}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|)\right)}}_{\text{Carrier}} \rightarrow \text{rotating field} \end{aligned}$$

In the right animations : $\alpha = 0, \beta = \frac{\pi}{2}$
(animations are plotted **without** the approximation)

$\bar{J} \neq 0$ - rotating field
 $\bar{J} = 0$ - standing field



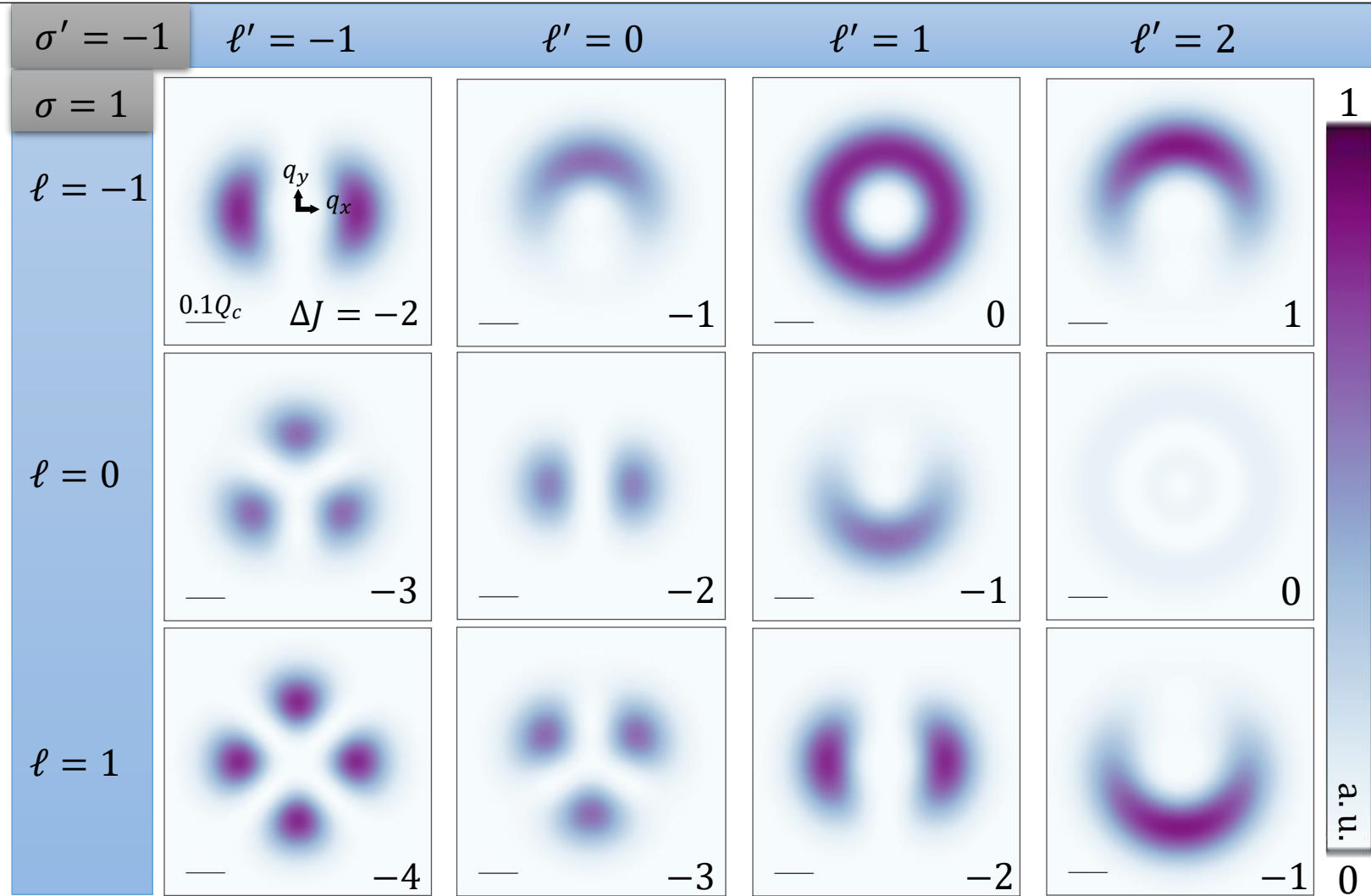
2. Longitudinal component (beating)

$$|\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta)|^2 = \left| G_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - i G_{|\ell|,|\ell'|}^-(q_{\parallel}; \beta) \sin\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right|^2$$

$$\approx \underbrace{\left| G_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right|^2}_{\text{Envelope}} \xrightarrow{\text{node structure}}$$

In the right figures : $\alpha = 0, \beta = \frac{\pi}{2}$
 (Figures are plotted **without** the approximation)

Number of nodes = $|\Delta J|$

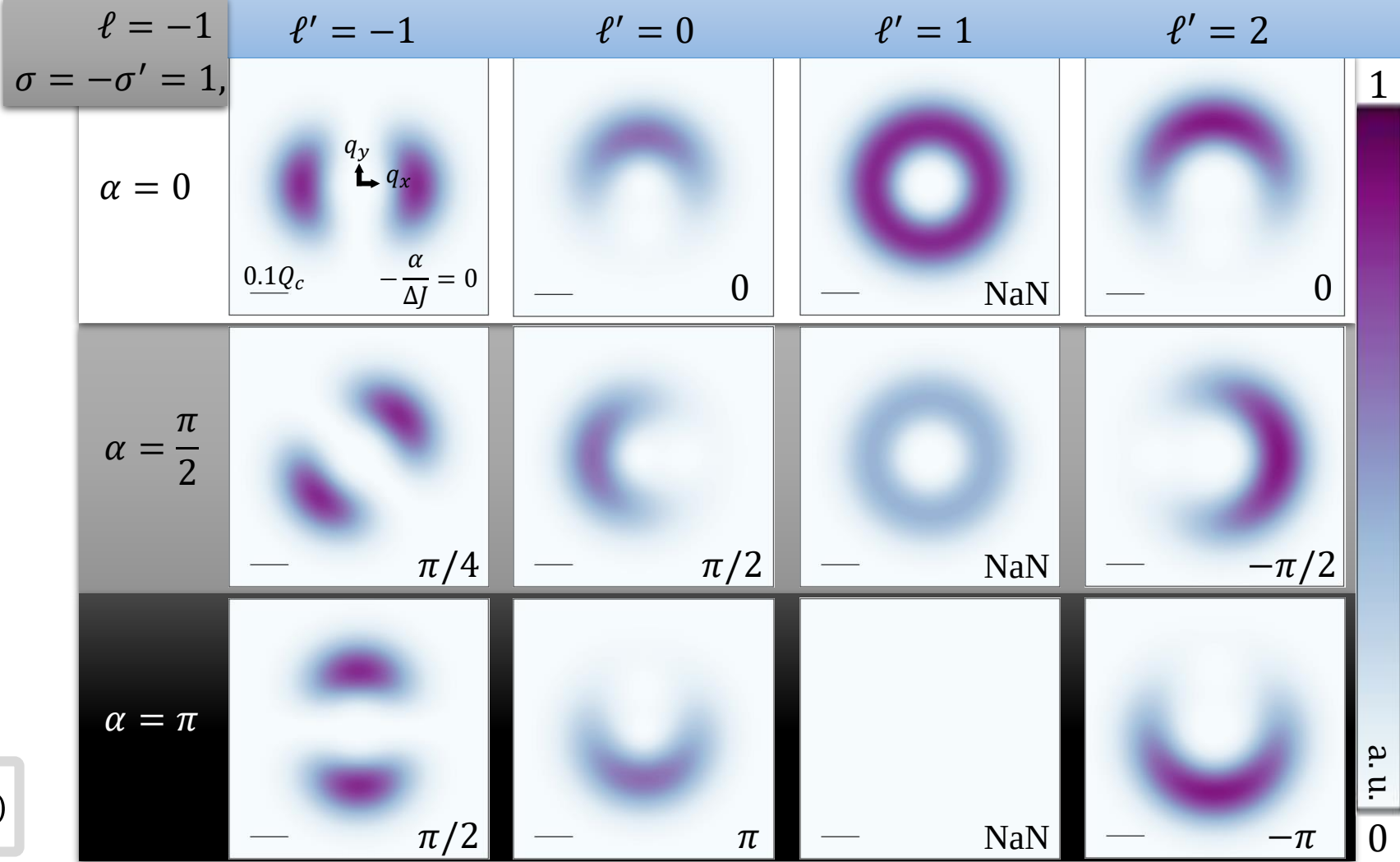


2. Longitudinal component (beating - relative phase α)

$$|\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta)|^2 = \left| G_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - i G_{|\ell|,|\ell'|}^-(q_{\parallel}; \beta) \sin\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right|^2$$

$$\approx \underbrace{\left| G_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right|^2}_{\text{Envelope}} \rightarrow \text{intensity change}$$

In the right figures : $\beta = \frac{\pi}{2}$
 (Figures are plotted **without** the approximation)

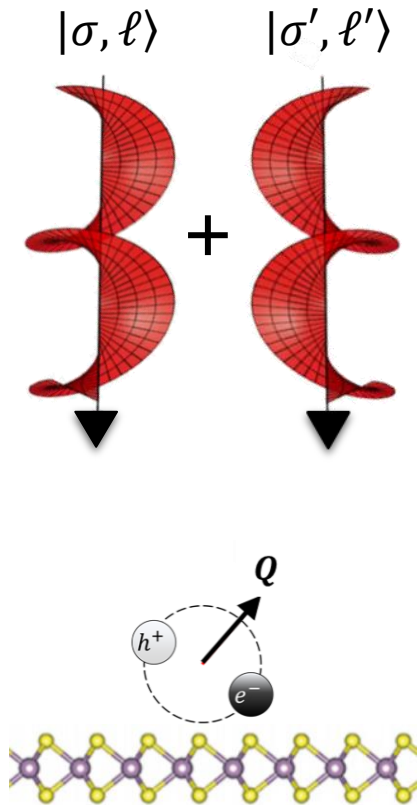


α change the intensity
 (rotate $-\frac{\alpha}{\Delta J}$ in counter-clockwise)

Relative phase : $\alpha_{|\ell|,|\ell'|} = \alpha - \frac{\pi}{2}(|\ell'| - |\ell|)$

The optical matrix element / transition rate of gray exciton

3. Transition rate (with approximation)



The transition rate is obtained as

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}) \right|^2 \rho(\hbar\omega = E_{S,Q}^X), \quad S = B+, B-, G$$

The optical matrix element is

$$\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{G,Q}(\mathbf{Q}, t; \alpha, \beta) = \frac{E_g}{2i\hbar} \left(\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{Q}, t; \alpha, \beta) \cdot \mathbf{D}^{G,Q*} \right) \text{ where } \mathbf{D}^{G,Q} = D^G \hat{\mathbf{z}}$$

$$\approx \underbrace{\frac{E_g}{2i\hbar} G_{|\ell|,|\ell'|}^+(Q; \beta) \cos\left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right)}_{\text{Envelope}} \times \underbrace{e^{i\left(\bar{J}\phi_Q - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|)\right)}}_{\text{Carrier}} \underbrace{(\hat{\mathbf{z}} \cdot D^G \hat{\mathbf{z}})}_{\text{Constant}}$$

node structure
adjust intensity
rotating

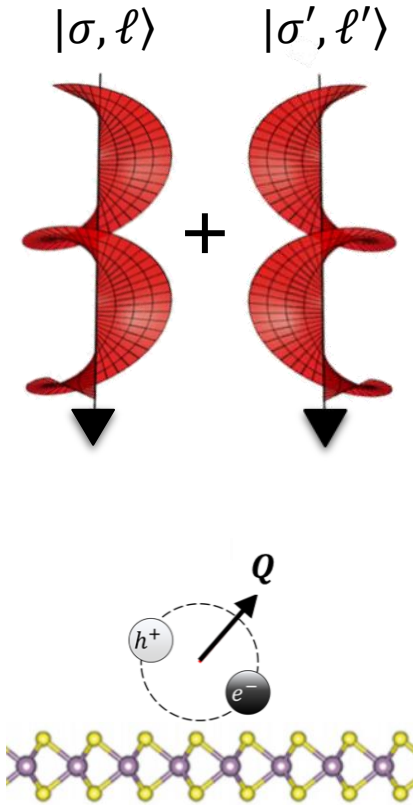
3. Transition rate (without approximation)

The transition rate is obtained as

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,\mathbf{Q}}(\mathbf{Q}) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,\mathbf{Q}}(\mathbf{Q}) \right|^2 \rho(\hbar\omega = E_{S,\mathbf{Q}}^X), \quad S = B+, B-, G$$

The optical matrix element is

$$\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{G,\mathbf{Q}}(\mathbf{Q}, t; \alpha, \beta) = \frac{E_g}{2i\hbar} \left(\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{Q}, t; \alpha, \beta) \cdot \mathbf{D}^{G,\mathbf{Q}*} \right) \text{ where } \mathbf{D}^{G,\mathbf{Q}} = D^G \hat{\mathbf{z}}$$

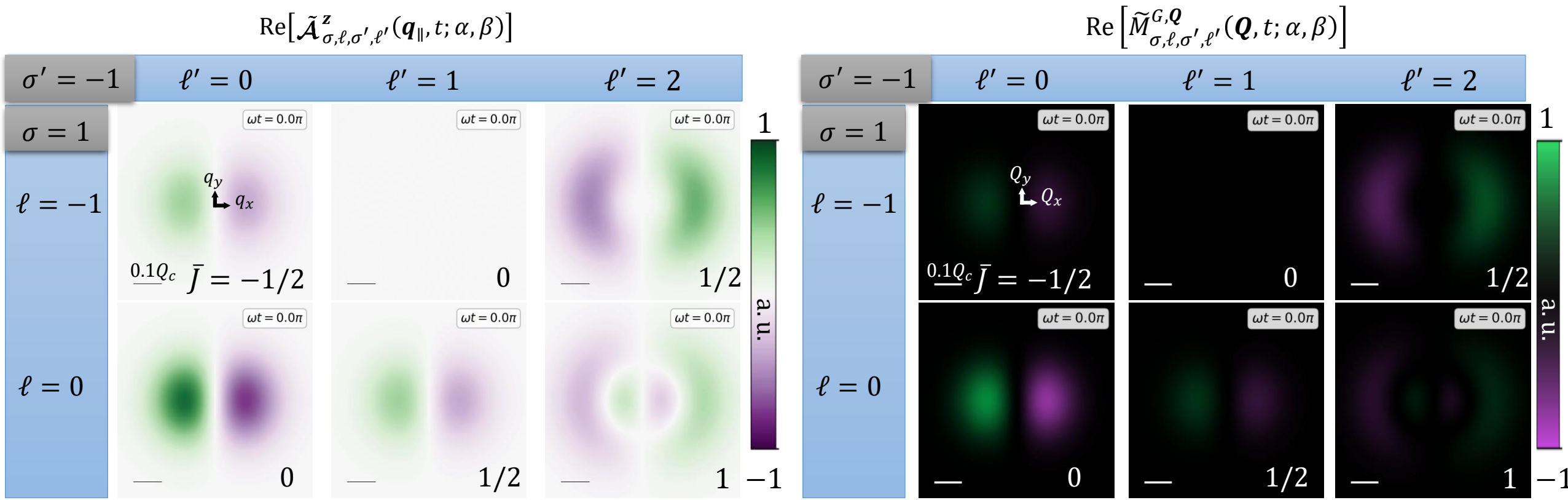


$$= \frac{E_g}{2i\hbar} \left[G_{|\ell|,|\ell'|}^+(Q; \beta) \cos\left(\frac{\Delta J}{2} \phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - \underbrace{iG_{|\ell|,|\ell'|}^-(Q; \beta) \sin\left(\frac{\Delta J}{2} \phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right)}_{\text{Previously neglected}} \right] \\ \times e^{i\left(\bar{J}\phi_{\mathbf{Q}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|)\right)} (\hat{\mathbf{z}} \cdot D^G \hat{\mathbf{z}})$$

3. Optical matrix element of gray exciton ($\bar{J} \neq 0$ - rotating ; $\bar{J} = 0$ - standing)

$$\begin{aligned} \text{Re} \left[\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{G,Q}(\mathbf{Q}, t; \alpha, \beta) \right] &= \text{Re} \left\{ \frac{E_g D^G}{2i\hbar} \left[G_{|\ell|,|\ell'|}^+(Q; \beta) \cos\left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - i G_{|\ell|,|\ell'|}^-(Q; \beta) \sin\left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \times e^{i(\bar{J}\phi_Q - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|))} \right\} \\ &\approx \underbrace{\text{Re} \left[\frac{E_g D^G}{2i\hbar} G_{|\ell|,|\ell'|}^+(Q; \beta) \cos\left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right]}_{\text{Envelope}} \underbrace{\times e^{i(\bar{J}\phi_Q - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|))}}_{\text{Carrier}} \longrightarrow \text{rotating field} \end{aligned}$$

In the below animations : $\alpha = 0, \beta = \frac{\pi}{2}$ (animations are plotted without the approximation)

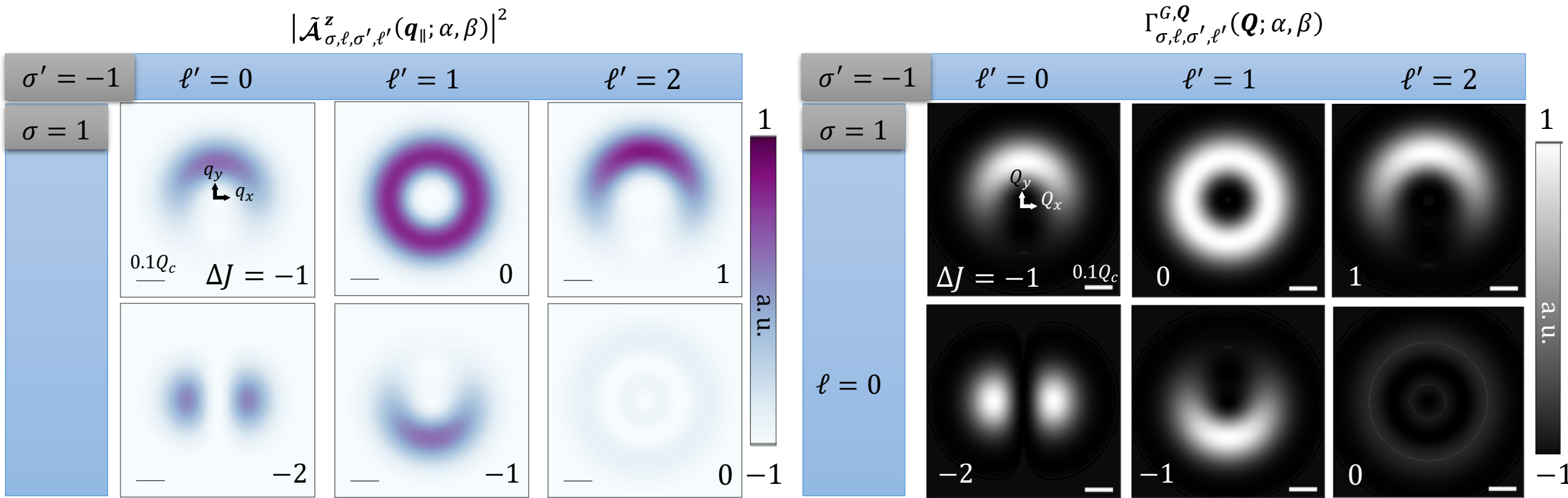


3. Transition rate of gray exciton (Number of nodes = $|\Delta J|$)

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{G,Q}(\mathbf{Q}; \alpha, \beta) \propto \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha, \beta) \right|^2 = \left| \frac{E_g D^G}{2i\hbar} \left[G_{|\ell|,|\ell'|}^+(Q; \beta) \cos\left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - i G_{|\ell|,|\ell'|}^-(Q; \beta) \sin\left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \right|^2$$

$$\approx \underbrace{\left| \frac{E_g D^G}{2i\hbar} G_{|\ell|,|\ell'|}^+(Q; \beta) \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right|^2}_{\text{Envelope}} \rightarrow \text{node structure}$$

In the below figures : $\alpha = 0, \beta = \frac{\pi}{2}$ (figures are plotted without the approximation)

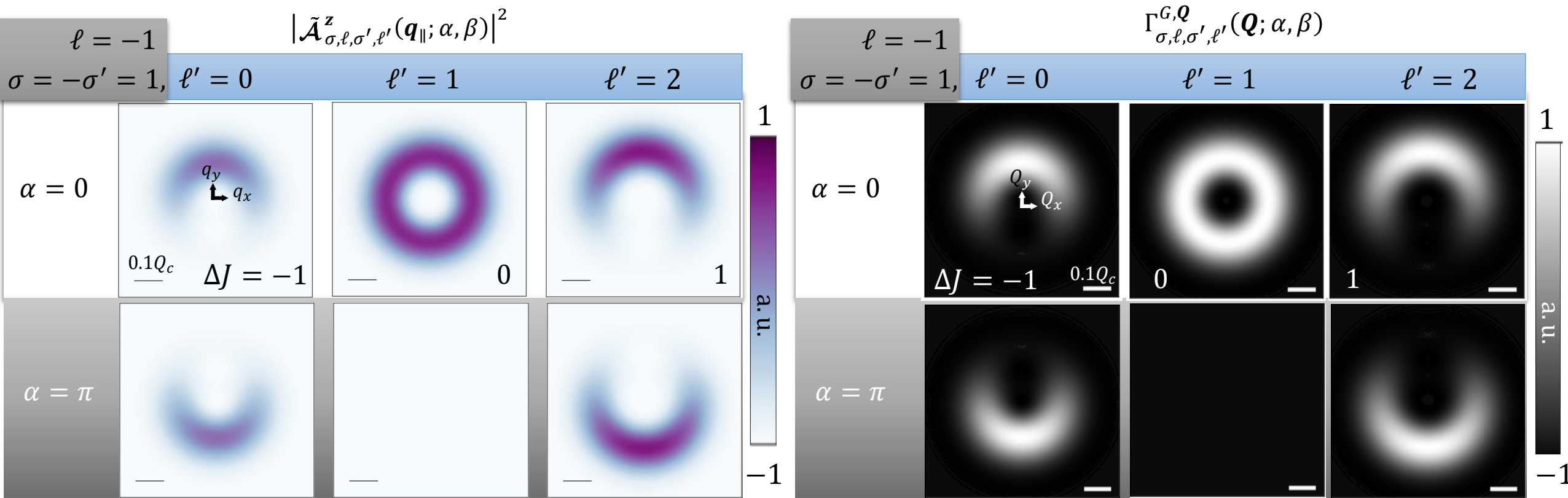


3. Transition rate of gray exciton (α adjust intensity)

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{G,Q}(\mathbf{Q}; \alpha, \beta) \propto \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha, \beta) \right|^2 = \left| \frac{E_g D^G}{2i\hbar} \left[G_{|\ell|,|\ell'|}^+(Q; \beta) \cos\left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - i G_{|\ell|,|\ell'|}^-(Q; \beta) \sin\left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \right|^2$$

$$\approx \underbrace{\left| \frac{E_g D^G}{2i\hbar} G_{|\ell|,|\ell'|}^+(Q; \beta) \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right|^2}_{\text{Envelope}} \longrightarrow \text{adjust intensity}$$

In the below figures : $\beta = \frac{\pi}{2}$ (figures are plotted without the approximation)



Transverse component of VVB

4. Signal additive mixing (2D)

Introduce opposite spin in $\hat{\mathbf{y}}$ axis

$$\begin{array}{ccc}
 \hat{x} + i\sigma\hat{y} & \xrightarrow[\substack{\Delta\sigma \equiv \sigma' - \sigma \\ \bar{\sigma} \equiv \frac{\sigma' + \sigma}{2}}]{} & (\hat{x} + \bar{\sigma}\hat{y}) - i\frac{\Delta\sigma}{2}\hat{y} \xrightarrow[\substack{\Delta\sigma = \pm 2 \\ \bar{\sigma} = 0}]{} (\hat{x} - i\frac{\Delta\sigma}{2}\hat{y}) \\
 \hat{x} + i\sigma'\hat{y} & \xrightarrow{} & (\hat{x} + \bar{\sigma}\hat{y}) + i\frac{\Delta\sigma}{2}\hat{y} \xrightarrow{} (\hat{x} + i\frac{\Delta\sigma}{2}\hat{y})
 \end{array}$$

Introducing opposite spin in $\hat{\mathbf{y}}$ axis change the “additive” mixing to “subtractive” mixing

Beating formulation :

In $\hat{\mathbf{x}}$ axis : $\text{Re}[\tilde{A}e^{i(k\phi - \omega t)} + \tilde{A}'e^{i(k'\phi - \omega t + \alpha)}]\hat{\mathbf{x}} \approx \text{Re}\left\{(|\tilde{A}| + |\tilde{A}'|)\cos\left(\frac{\Delta k}{2}\phi + \frac{\alpha'}{2}\right)e^{i\left(\bar{k}\phi - \omega t + \frac{\alpha}{2} + \frac{1}{2}(\text{Arg}[\tilde{A}'] + \text{Arg}[\tilde{A}])\right)}\right\}\hat{\mathbf{x}}$

In $\hat{\mathbf{y}}$ axis : $\text{Re}\left[-i\frac{\Delta\sigma}{2}\tilde{A}e^{i(k\phi - \omega t)} + i\frac{\Delta\sigma}{2}\tilde{A}'e^{i(k'\phi - \omega t + \alpha)}\right]\hat{\mathbf{y}} \approx \text{Re}\left\{-\frac{\Delta\sigma}{2}(|\tilde{A}| + |\tilde{A}'|)\sin\left(\frac{\Delta k}{2}\phi + \frac{\alpha'}{2}\right)e^{i\left(\bar{k}\phi - \omega t + \frac{\alpha}{2} + \frac{1}{2}(\text{Arg}[\tilde{A}'] + \text{Arg}[\tilde{A}])\right)}\right\}\hat{\mathbf{y}}$

In 2D:

$$\text{Re}\left[\tilde{A}e^{i(k\phi - \omega t)}\left(\hat{x} - i\frac{\Delta\sigma}{2}\hat{y}\right) + \tilde{A}'e^{i(k'\phi - \omega t + \alpha)}\left(\hat{x} + i\frac{\Delta\sigma}{2}\hat{y}\right)\right] \approx \underbrace{\text{Re}\left\{(|\tilde{A}| + |\tilde{A}'|)\left[\cos\left(\frac{\Delta k}{2}\phi + \frac{\alpha'}{2}\right)\hat{x} - \frac{\Delta\sigma}{2}\sin\left(\frac{\Delta k}{2}\phi + \frac{\alpha'}{2}\right)\hat{y}\right]\right\}}_{\text{Envelope in 2D (polarization)}} \underbrace{e^{i\left(\bar{k}\phi - \omega t + \frac{\alpha}{2} + \frac{1}{2}(\text{Arg}[\tilde{A}'] + \text{Arg}[\tilde{A}])\right)}}_{\text{Carrier}}$$

2. Analogy to beating formulation (with approximation)

Beating formulation :

$$\begin{aligned} \text{Sum} &= \text{Re} \left[\tilde{A} e^{i(k\phi - \omega t)} \left(\hat{\mathbf{x}} - i \frac{\Delta\sigma}{2} \hat{\mathbf{y}} \right) + \tilde{A}' e^{i(k'\phi - \omega t + \alpha)} \left(\hat{\mathbf{x}} + i \frac{\Delta\sigma}{2} \hat{\mathbf{y}} \right) \right] \\ &\approx \text{Re} \left\{ \underbrace{(|\tilde{A}| + |\tilde{A}'|) \left[\cos \left(\frac{\Delta k}{2} \phi + \frac{\alpha'}{2} \right) \hat{\mathbf{x}} - \frac{\Delta\sigma}{2} \sin \left(\frac{\Delta k}{2} \phi + \frac{\alpha'}{2} \right) \hat{\mathbf{y}} \right]}_{\text{Envelope in 2D (polarization)}} \underbrace{e^{i \left(\bar{k}\phi - \omega t + \frac{\alpha}{2} + \frac{1}{2}(\text{Arg}[\tilde{A}'] + \text{Arg}[\tilde{A}]) \right)}}_{\text{Carrier}} \right\} \end{aligned}$$

Transverse
component of VVB :

$$\begin{aligned} \tilde{\mathcal{A}}_{\sigma, \ell, \sigma', \ell'}^{\parallel}(\mathbf{q}_{\parallel}, t; \alpha, \beta) &= \frac{1}{\sqrt{2}} \cos \left(\frac{\beta}{2} \right) \tilde{F}_{|\ell|}(q_{\parallel}) e^{i(\ell\phi_{q_{\parallel}} - \omega t)} \left(\hat{\mathbf{x}} - i \frac{\Delta\sigma}{2} \hat{\mathbf{y}} \right) + \frac{1}{\sqrt{2}} \sin \left(\frac{\beta}{2} \right) \tilde{F}_{|\ell'|}(q_{\parallel}) e^{i(\ell'\phi_{q_{\parallel}} - \omega t + \alpha)} \left(\hat{\mathbf{x}} - i \frac{\Delta\sigma}{2} \hat{\mathbf{y}} \right) \\ &\approx \underbrace{H_{|\ell|, |\ell'|}^{+}(q_{\parallel}; \beta) \left[\cos \left(\frac{\Delta\ell}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{x}} - \frac{\Delta\sigma}{2} \sin \left(\frac{\Delta\ell}{2} \phi_{q_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{y}} \right]}_{\text{Polarization}} \times \underbrace{e^{i \left(\bar{\ell}\phi_{q_{\parallel}} + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|) \right)}}_{\text{Carrier}} \end{aligned}$$

Winding of light
rotate polarization
rotating field

Definition :

$$\begin{aligned} |\tilde{A}| \pm |\tilde{A}'| &\longrightarrow H_{|\ell|, |\ell'|}^{\pm}(q_{\parallel}; \beta) \equiv \left| \frac{1}{\sqrt{2}} \cos \left(\frac{\beta}{2} \right) \tilde{F}_{|\ell|}(q_{\parallel}) \right| \pm \left| \frac{1}{\sqrt{2}} \sin \left(\frac{\beta}{2} \right) \tilde{F}_{|\ell'|}(q_{\parallel}) \right| \\ \Delta k \equiv k' - k \text{ and } \bar{k} \equiv \frac{k' + k}{2} &\longrightarrow \Delta\ell \equiv \ell' - \ell \text{ and } \bar{\ell} \equiv \frac{\ell' + \ell}{2} \\ \alpha' \equiv \alpha + \text{Arg}[\tilde{A}'] - \text{Arg}[\tilde{A}] &\longrightarrow \alpha_{|\ell|, |\ell'|} \equiv \alpha + \text{Arg}[\tilde{F}_{|\ell'|}(q_{\parallel})] - \text{Arg}[\tilde{F}_{|\ell|}(q_{\parallel})] = \alpha - \frac{\pi}{2}(|\ell'| - |\ell|) \end{aligned}$$

2. Analogy to beating formulation (without approximation)

Beating formulation :

Without assuming $|\tilde{A}| + |\tilde{A}'| \gg |\tilde{A}| - |\tilde{A}'|$

$$\text{Sum} = \text{Re} \left\{ \left[(|\tilde{A}| + |\tilde{A}'|) \left[\cos\left(\frac{\Delta k}{2} \phi + \frac{\alpha'}{2}\right) \hat{x} - \frac{\Delta \sigma}{2} \sin\left(\frac{\Delta k}{2} \phi + \frac{\alpha'}{2}\right) \hat{y} \right] - \underbrace{i(|\tilde{A}| - |\tilde{A}'|) \left[\sin\left(\frac{\Delta k}{2} \phi + \frac{\alpha'}{2}\right) \hat{x} + \frac{\Delta \sigma}{2} \cos\left(\frac{\Delta k}{2} \phi + \frac{\alpha'}{2}\right) \hat{y} \right]}_{\text{Previously neglected}} \right] e^{i\left(\bar{k}\phi - \omega t + \frac{\alpha}{2} + \frac{1}{2}(\text{Arg}[\tilde{A}'] + \text{Arg}[\tilde{A}])\right)} \right\}$$

Transverse
component of VVB :

Without assuming $H_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \gg H_{|\ell|,|\ell'|}^-(q_{\parallel}; \beta)$

$$\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}, t; \alpha, \beta) = \left\{ H_{|\ell|,|\ell'|}^+(q_{\parallel}; \beta) \left[\cos\left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \hat{x} - \frac{\Delta \sigma}{2} \sin\left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \hat{y} \right] - \right. \\ \left. \underbrace{i H_{|\ell|,|\ell'|}^-(q_{\parallel}; \beta) \left[\sin\left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \hat{x} + \frac{\Delta \sigma}{2} \cos\left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \hat{y} \right]}_{\text{Previously neglected}} \right\} \times e^{i\left(\bar{\ell}\phi_{\mathbf{q}_{\parallel}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|)\right)}$$

4. Transverse component (rotating/standing)

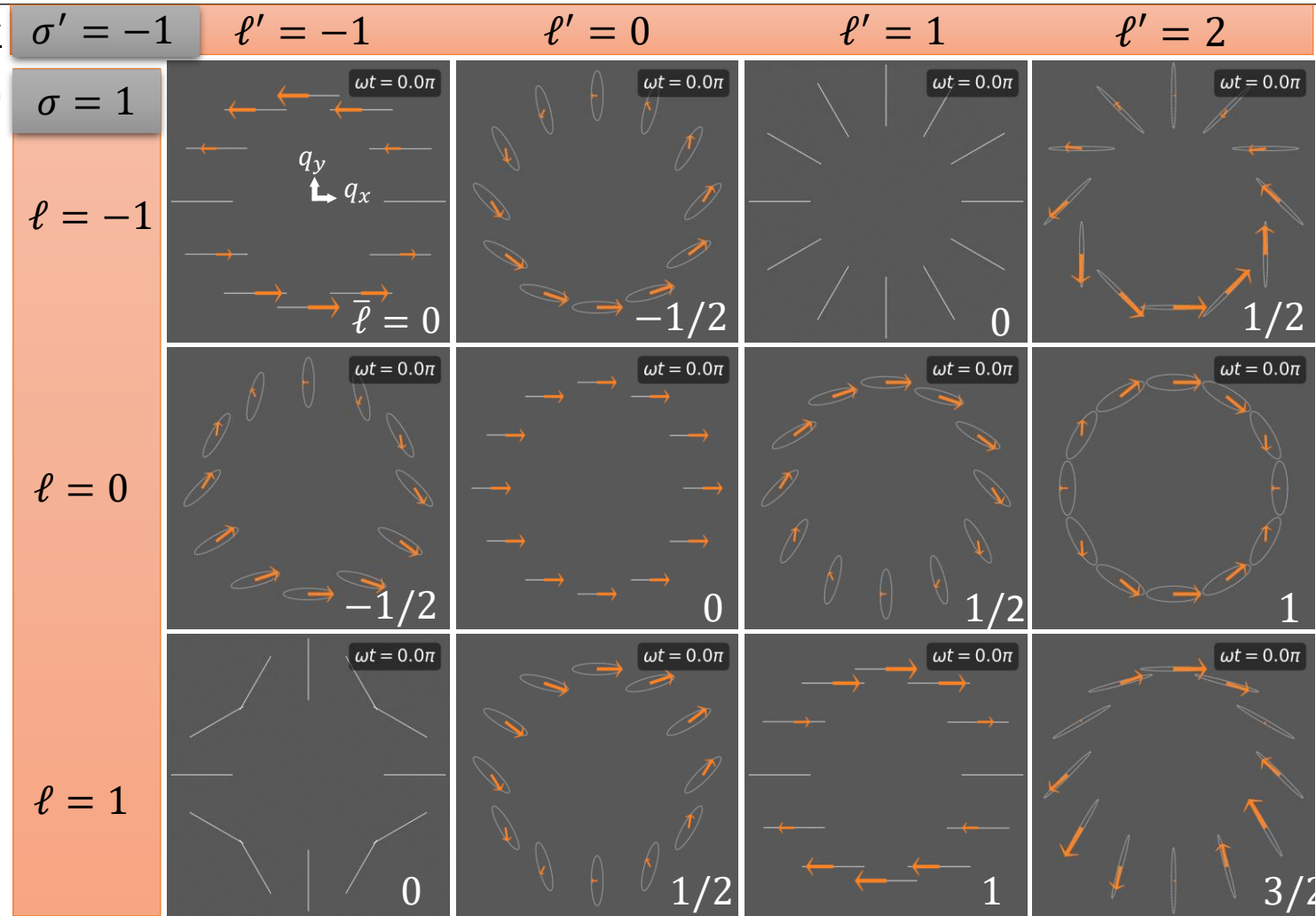
$$\text{Re} \left[\tilde{\mathcal{A}}_{\sigma, \ell, \sigma', \ell'}^{\parallel}(\mathbf{q}_{\parallel}, t; \alpha, \beta) \right] \approx \underbrace{\text{Re} \left\{ H_{|\ell|, |\ell'|}^{+}(\mathbf{q}_{\parallel}; \beta) \left[\cos \left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{x}} - \frac{\Delta \sigma}{2} \sin \left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{y}} \right] \right\}}_{\text{Polarization}} \times \underbrace{e^{i \left(\bar{\ell} \phi_{\mathbf{q}_{\parallel}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4} (|\ell'| + |\ell|) \right)}}_{\text{Carrier}} \xrightarrow{\text{rotating field}}$$

In the right animations : $\alpha = 0, \beta = \frac{\pi}{2}$
(animations are plotted **without** the approximation)

Focus on instantons polarization (orange arrow)

$\bar{\ell} \neq 0$ - rotating field

$\bar{\ell} = 0$ - standing field (in phase)



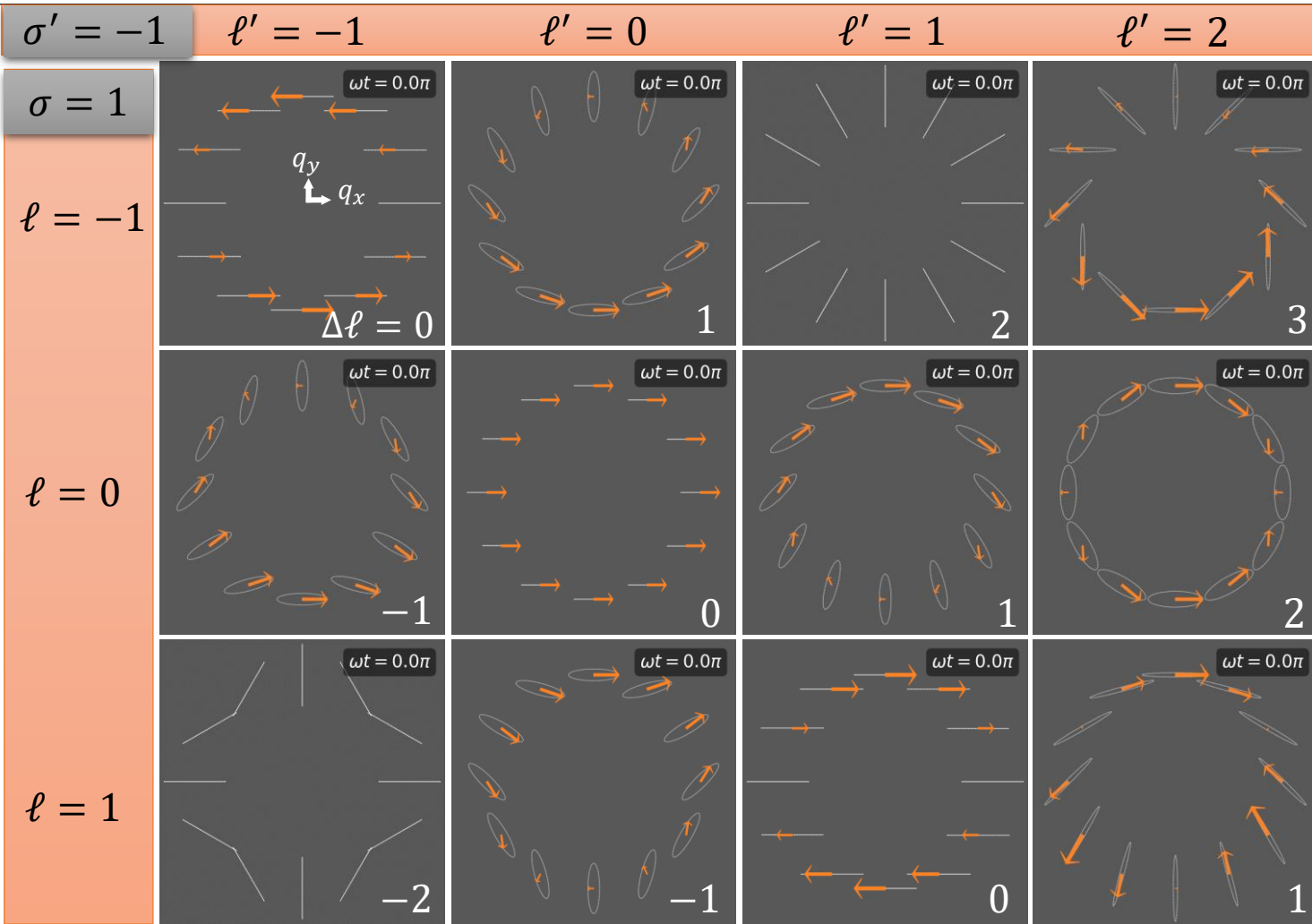
4. Transverse component (beating)

$$\text{Re} \left[\tilde{\mathcal{A}}_{\sigma, \ell, \sigma', \ell'}^{\parallel}(\mathbf{q}_{\parallel}, t; \alpha, \beta) \right] \approx \underbrace{\text{Re} \left\{ H_{|\ell|, |\ell'|}^{+}(\mathbf{q}_{\parallel}; \beta) \left[\cos \left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{x}} - \frac{\Delta \sigma}{2} \sin \left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{y}} \right] \right\}}_{\text{Polarization}} \times \underbrace{e^{i \left(\bar{\ell} \phi_{\mathbf{q}_{\parallel}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4} (|\ell'| + |\ell|) \right)}}_{\text{Carrier}}$$

winding of light

In the right animations : $\alpha = 0, \beta = \frac{\pi}{2}$
(animations are plotted **without** the approximation)

winding of light = $\Delta \ell$
(polarization, white trajectory)



4. Transverse component (beating - relative phase α)

$$\text{Re} \left[\tilde{\mathcal{A}}_{\sigma, \ell, \sigma', \ell'}^{\parallel}(\mathbf{q}_{\parallel}, t; \alpha, \beta) \right] \approx \underbrace{\text{Re} \left\{ H_{|\ell|, |\ell'|}^{+}(\mathbf{q}_{\parallel}; \beta) \left[\cos \left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{x}} - \frac{\Delta \sigma}{2} \sin \left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{y}} \right] \right\}}_{\text{Polarization}} \times \underbrace{e^{i \left(\bar{\ell} \phi_{\mathbf{q}_{\parallel}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4} (|\ell'| + |\ell|) \right)}}_{\text{Carrier}}$$

rotate polarization

In the right animations : $\beta = \frac{\pi}{2}$
(animations are plotted **without** the approximation)

$\ell = -1$
 $\sigma = -\sigma' = 1,$

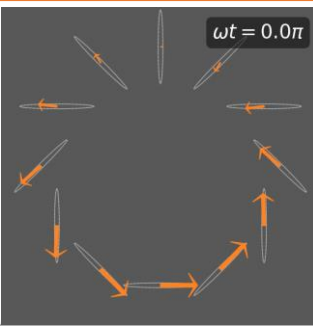
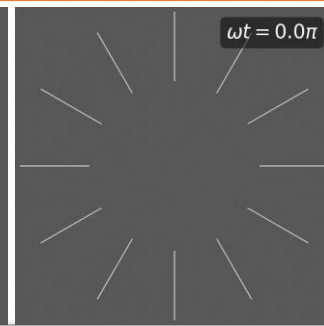
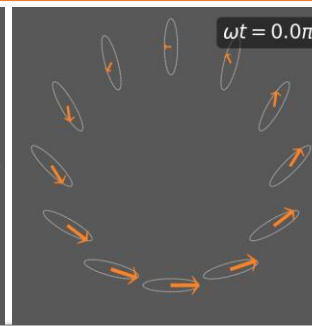
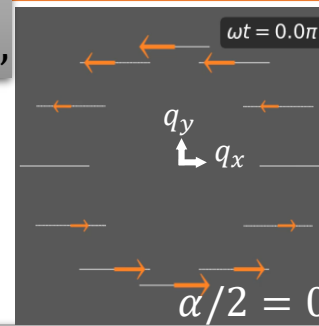
$\ell' = -1$

$\ell' = 0$

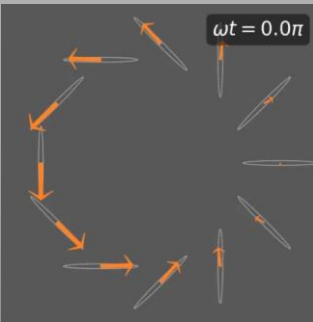
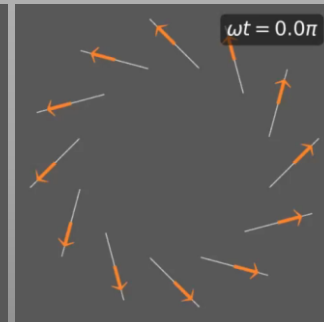
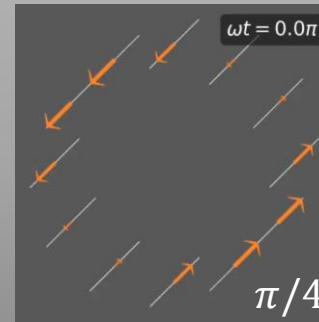
$\ell' = 1$

$\ell' = 2$

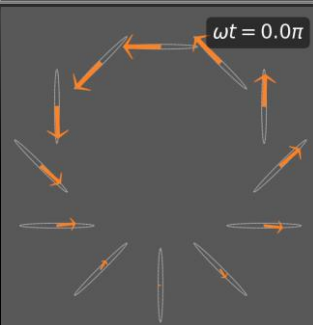
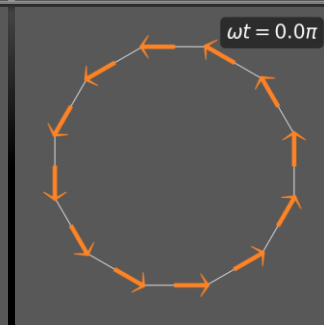
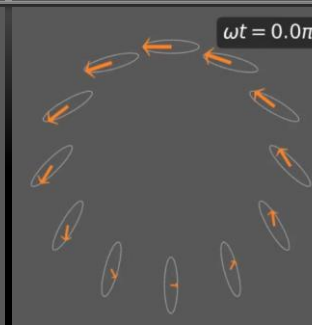
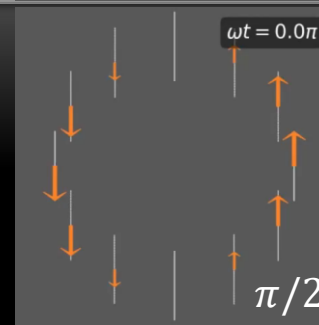
$\alpha = 0$



$\alpha = \frac{\pi}{2}$



$\alpha = \pi$



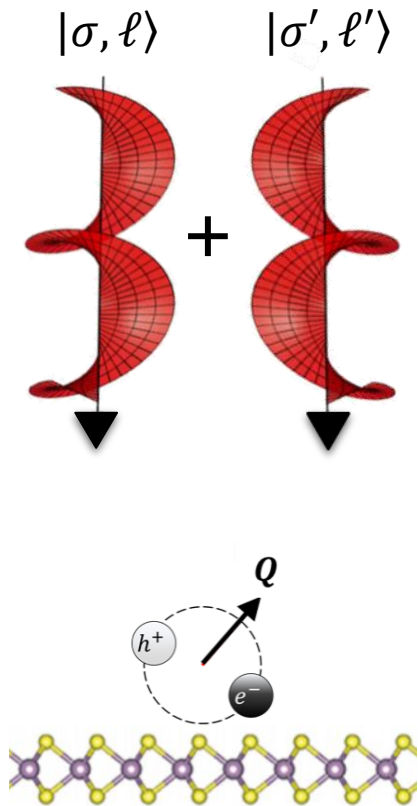
α rotate each polarization

(polarization, white trajectory,
will rotated by α in counter-clockwise)

Relative phase : $\alpha_{|\ell|, |\ell'|} = \alpha - \frac{\pi}{2} (|\ell'| - |\ell|)$

The optical matrix element / transition rate of bright exciton

5. Transition rate



The transition rate is obtained as

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}) \right|^2 \rho(\hbar\omega = E_{S,Q}^X),$$

The optical matrix element is

$$\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{B\pm,Q}(\mathbf{Q}, t; \alpha, \beta) = \frac{E_g}{2i\hbar} \left(\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(Q, t; \alpha, \beta) \cdot \mathbf{D}^{B\pm,Q*} \right) \quad \text{where} \quad \begin{aligned} \mathbf{D}^{B+,Q} &= D_B^X (\cos \phi_Q \hat{x} + \sin \phi_Q \hat{y}) \\ \mathbf{D}^{B-,Q} &= iD_B^X (-\sin \phi_Q \hat{x} + \cos \phi_Q \hat{y}) \end{aligned}$$

$$\approx \frac{E_g}{2i\hbar} H_{|\ell|,|\ell'|}^+(Q; \beta) \times$$

$$\left\{ \left[\cos \left(\frac{\Delta\ell}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2} \right) \hat{x} - \frac{\Delta\sigma}{2} \sin \left(\frac{\Delta\ell}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2} \right) \hat{y} \right] \cdot \mathbf{D}^{B\pm,Q*} \right\}$$

Product of polarization and $\mathbf{D}^{B\pm,Q*}$

$$\times e^{i\left(\bar{\ell}\phi_Q - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|)\right)}$$

5. Product of polarization and $\mathbf{D}^{B\pm, \mathbf{Q}*}$

1. Rewrite the product of polarization and $\mathbf{D}^{B\pm, \mathbf{Q}*}$

$$\left[\cos \left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{x}} - \frac{\Delta \sigma}{2} \sin \left(\frac{\Delta \ell}{2} \phi_{\mathbf{q}_{\parallel}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \hat{\mathbf{y}} \right] \cdot \mathbf{D}^{B\pm, \mathbf{Q}*} = \left[\cos \theta \hat{\mathbf{x}} - \frac{\Delta \sigma}{2} \sin \theta \hat{\mathbf{y}} \right] \cdot \mathbf{D}^{B\pm, \mathbf{Q}*}$$

2. Consider transition dipole $\mathbf{D}^{B+, \mathbf{Q}} = D^B (\cos \phi_{\mathbf{Q}} \hat{\mathbf{x}} + \sin \phi_{\mathbf{Q}} \hat{\mathbf{y}})$ and $\mathbf{D}^{B-, \mathbf{Q}} = iD^B (-\sin \phi_{\mathbf{Q}} \hat{\mathbf{x}} + \cos \phi_{\mathbf{Q}} \hat{\mathbf{y}})$

$$\begin{aligned} \left[\cos \theta \hat{\mathbf{x}} - \frac{\Delta \sigma}{2} \sin \theta \hat{\mathbf{y}} \right] \cdot \mathbf{D}^{B+, \mathbf{Q}*} &= D^B \left[\cos \phi_{\mathbf{Q}} \cos \theta - \frac{\Delta \sigma}{2} \sin \phi_{\mathbf{Q}} \sin \theta \right] \\ &= \frac{D^B}{2} \left[\left(1 - \frac{\Delta \sigma}{2} \right) \cos(\theta - \phi_{\mathbf{Q}}) + \left(1 + \frac{\Delta \sigma}{2} \right) \cos(\theta + \phi_{\mathbf{Q}}) \right] \end{aligned}$$

$$\begin{aligned} \left[\cos \theta \hat{\mathbf{x}} - \frac{\Delta \sigma}{2} \sin \theta \hat{\mathbf{y}} \right] \cdot \mathbf{D}^{B-, \mathbf{Q}*} &= -iD^B \left[-\sin \phi_{\mathbf{Q}} \cos \theta - \frac{\Delta \sigma}{2} \cos \phi_{\mathbf{Q}} \sin \theta \right] \\ &= \frac{-iD^B}{2} \left[\left(1 - \frac{\Delta \sigma}{2} \right) \sin(\theta - \phi_{\mathbf{Q}}) - \left(1 + \frac{\Delta \sigma}{2} \right) \sin(\theta + \phi_{\mathbf{Q}}) \right] \end{aligned}$$

Beating wavenumber is changed

winding of light winding number intensity change

$$\begin{aligned} \tilde{M}_{\sigma, \ell, \sigma', \ell'}^{B+, \mathbf{Q}}(\mathbf{Q}, t; \alpha, \beta) &\approx \frac{E_g D^B}{4i\hbar} H_{|\ell|, |\ell'|}^+(Q; \beta) \left[\left(1 - \frac{\Delta \sigma}{2} \right) \cos \left(\frac{\Delta \ell - 2}{2} \phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) + \left(1 + \frac{\Delta \sigma}{2} \right) \cos \left(\frac{\Delta \ell + 2}{2} \phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \right] e^{i(\bar{\ell} \phi_{\mathbf{Q}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|))} \\ \tilde{M}_{\sigma, \ell, \sigma', \ell'}^{B-, \mathbf{Q}}(\mathbf{Q}, t; \alpha, \beta) &\approx \frac{-iE_g D^B}{4i\hbar} H_{|\ell|, |\ell'|}^+(Q; \beta) \left[\left(1 - \frac{\Delta \sigma}{2} \right) \sin \left(\frac{\Delta \ell - 2}{2} \phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) - \left(1 + \frac{\Delta \sigma}{2} \right) \sin \left(\frac{\Delta \ell + 2}{2} \phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|, |\ell'|}}{2} \right) \right] e^{i(\bar{\ell} \phi_{\mathbf{Q}} - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|))} \end{aligned}$$

Envelope Carrier

5. Optical matrix element of upper bright exciton (rotating/standing)

$$\text{Re} \left[\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{B+,Q}(\mathbf{Q}, t; \alpha, \beta) \right] \approx \underbrace{\text{Re} \left\{ \frac{E_g D^B}{4i\hbar} H_{|\ell|,|\ell'|}^+(Q; \beta) \left[\left(1 - \frac{\Delta\sigma}{2}\right) \cos\left(\frac{\Delta\ell - 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \cos\left(\frac{\Delta\ell + 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \right\}}_{\text{Envelope}} \underbrace{e^{i\left(\bar{\ell}\phi_Q - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|)\right)}}_{\text{Carrier rotating}}$$

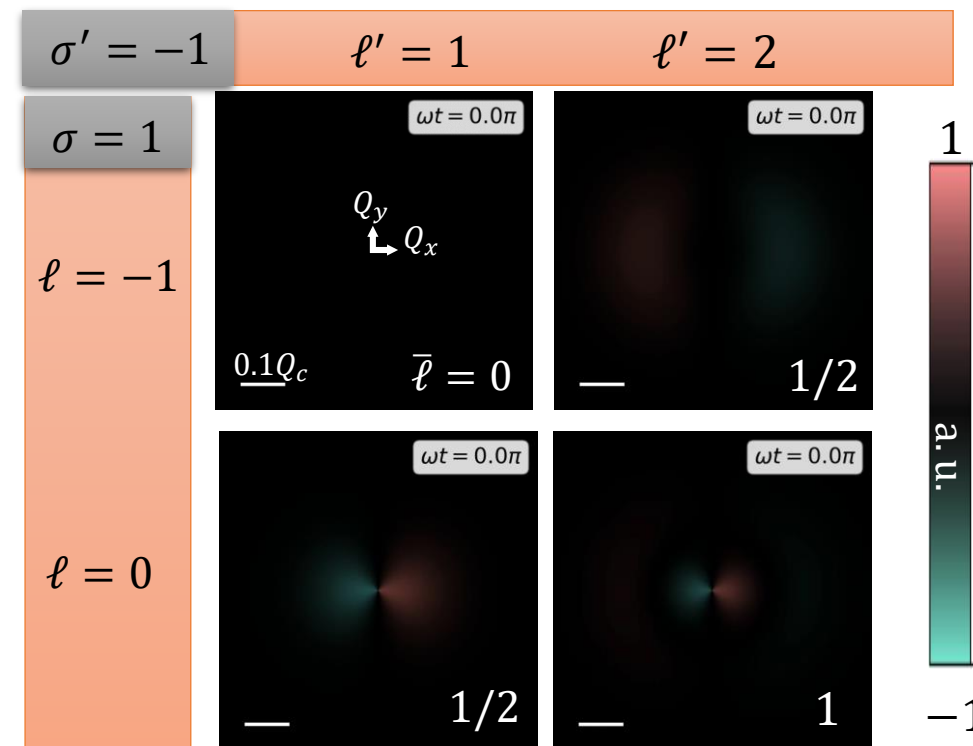
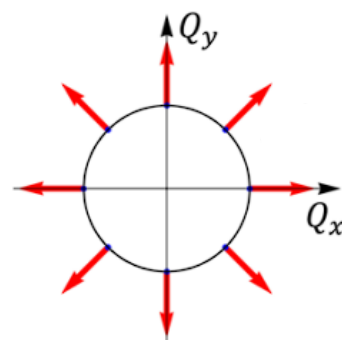
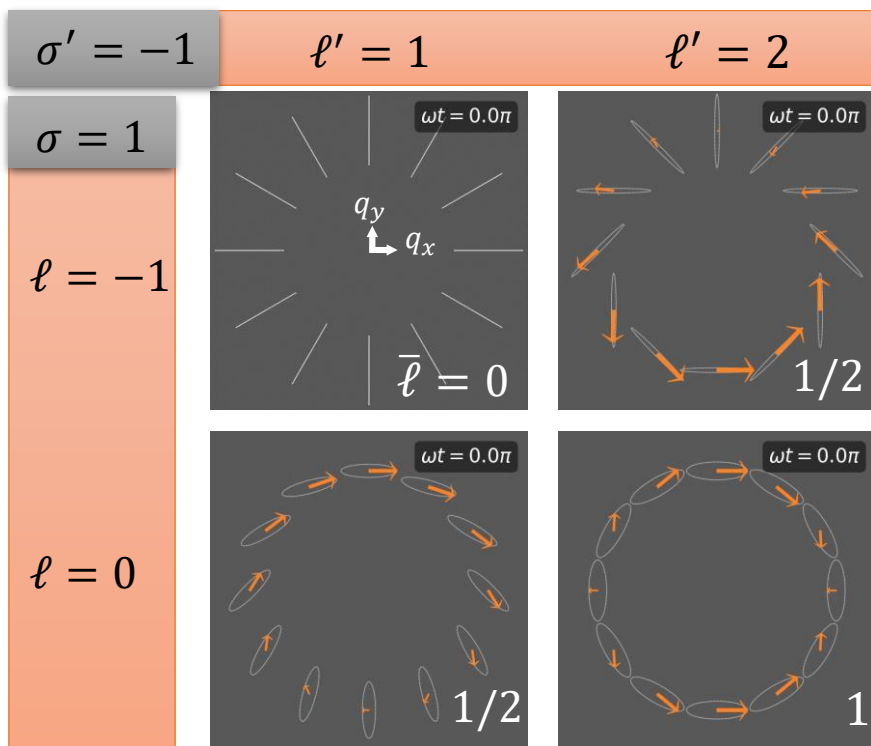
$\bar{\ell} \neq 0$ - rotating ; $\bar{\ell} = 0$ - standing

In the below figures : $\alpha = 0, \beta = \frac{\pi}{2}, \Delta\sigma = -2$ (animations are plotted **without** the approximation)

$$\text{Re} \left[\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}, t; \alpha, \beta) \right]$$

$$D^{B+,Q}$$

$$\text{Re} \left[\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{B+,Q}(\mathbf{Q}, t; \alpha, \beta) \right]$$



5. Transition rate of upper bright exciton (beating)

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{B+,Q}(\mathbf{Q}; \alpha, \beta) \propto \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{B+,Q}(\mathbf{Q}; \alpha, \beta) \right|^2 \approx \underbrace{\left| \frac{E_g D^B}{4i\hbar} H_{|\ell|,|\ell'|}^+(\mathbf{Q}; \beta) \left[\left(1 - \frac{\Delta\sigma}{2}\right) \cos\left(\frac{\Delta\ell - 2}{2} \phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \cos\left(\frac{\Delta\ell + 2}{2} \phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \right|^2}_{\text{node structure Envelope node structure}}$$

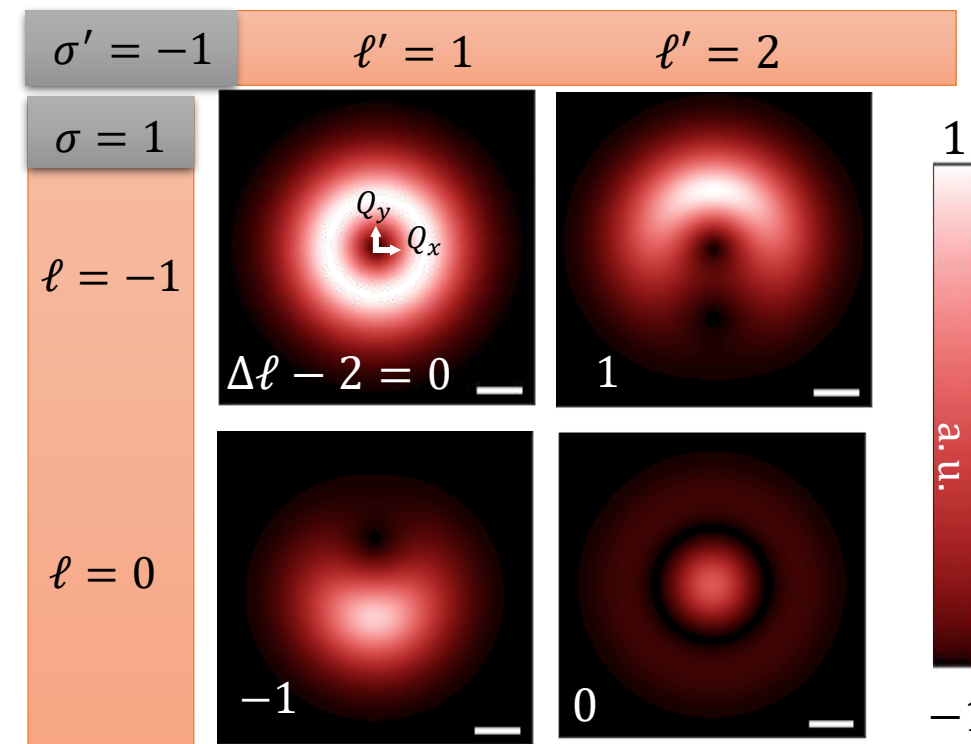
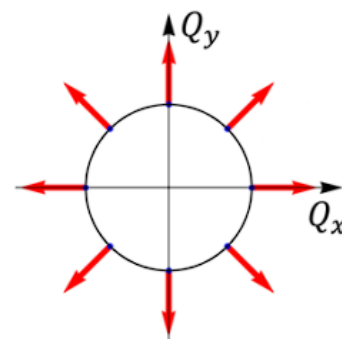
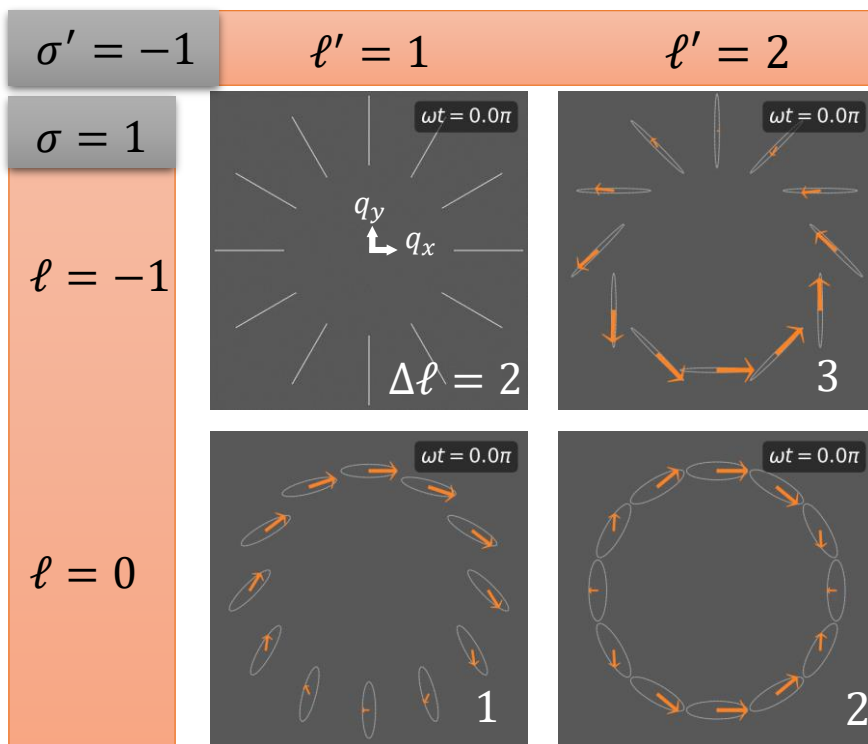
Number of nodes = $\Delta\ell - 2$ (for $\Delta\sigma = -2$)

In the below figures : $\alpha = 0, \beta = \frac{\pi}{2}, \Delta\sigma = -2$

$\text{Re} \left[\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}, t; \alpha, \beta) \right]$

$D^{B+,Q}$

$\Gamma_{\sigma,\ell,\sigma',\ell'}^{B+,Q}(\mathbf{Q}; \alpha, \beta)$



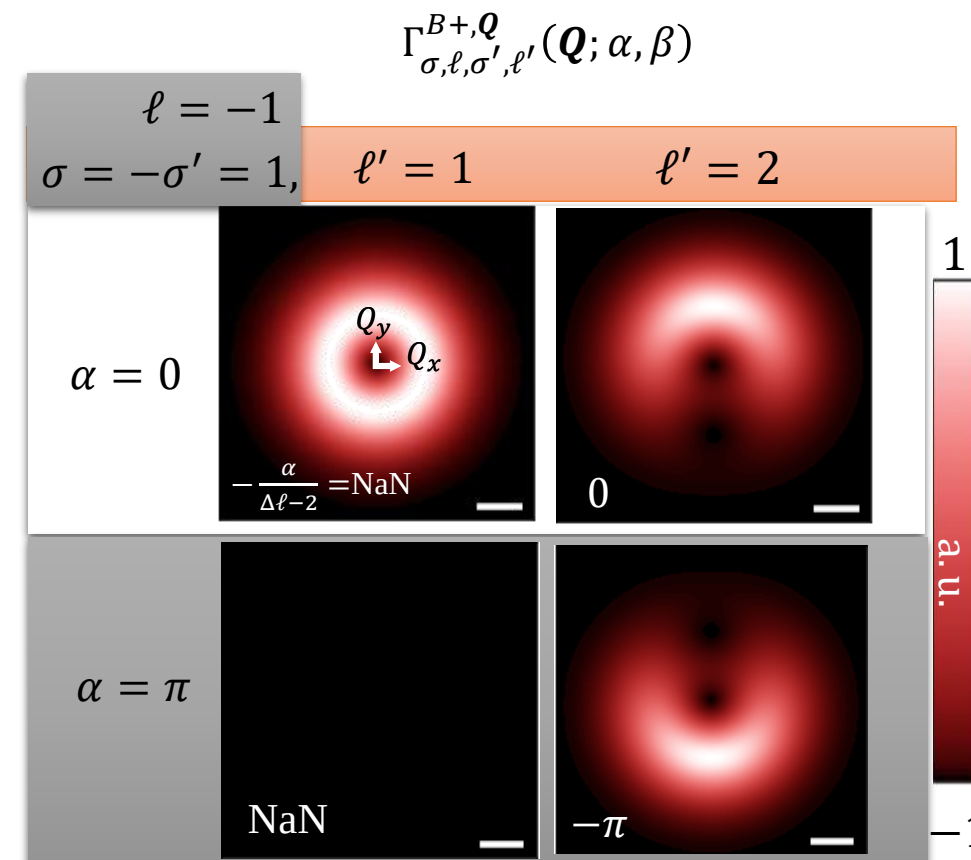
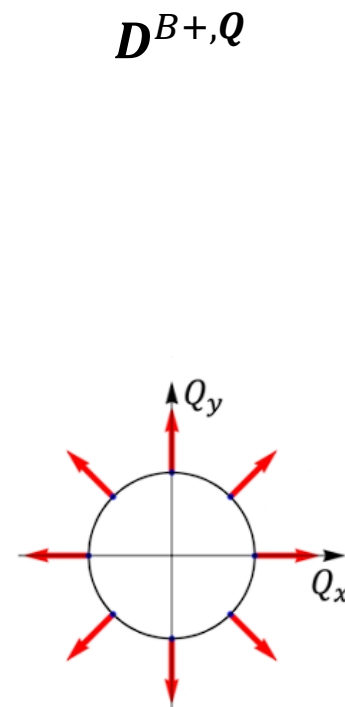
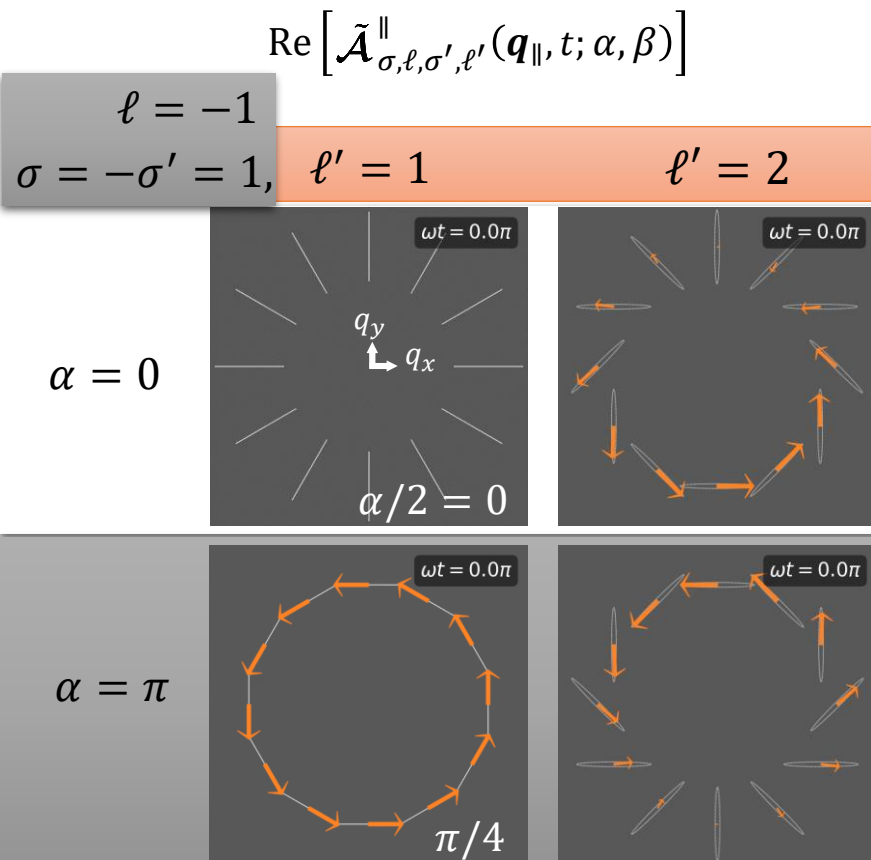
5. Transition rate of upper bright exciton (beating - relative phase α)

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{B+,Q}(\mathbf{Q}; \alpha, \beta) \propto \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{B+,Q}(\mathbf{Q}; \alpha, \beta) \right|^2 \approx \underbrace{\left| \frac{E_g D^B}{4i\hbar} H_{|\ell|,|\ell'|}^+(Q; \beta) \right|}_{\text{intensity change}} \underbrace{\left[\left(1 - \frac{\Delta\sigma}{2} \right) \cos\left(\frac{\Delta\ell - 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2} \right) + \left(1 + \frac{\Delta\sigma}{2} \right) \cos\left(\frac{\Delta\ell + 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2} \right) \right]}_{\text{Envelope}}^2$$

intensity change Envelope intensity change

α change the intensity

In the below figures : $\alpha = 0, \beta = \frac{\pi}{2}, \Delta\sigma = -2$



5. Optical matrix element of lower bright exciton (rotating/standing)

$$\text{Re} \left[\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{B-,Q}(\mathbf{Q}, t; \alpha, \beta) \right] \approx \underbrace{\text{Re} \left\{ \frac{-iE_g D^B}{4i\hbar} H_{|\ell|,|\ell'|}^+(Q; \beta) \left[\left(1 - \frac{\Delta\sigma}{2}\right) \sin\left(\frac{\Delta\ell - 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - \left(1 + \frac{\Delta\sigma}{2}\right) \sin\left(\frac{\Delta\ell + 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \right\}}_{\text{Envelope}} \underbrace{e^{i(\bar{\ell}\phi_Q - \omega t + \frac{\alpha}{2} - \frac{\pi}{4}(|\ell'| + |\ell|))}}_{\text{Carrier rotating}}$$

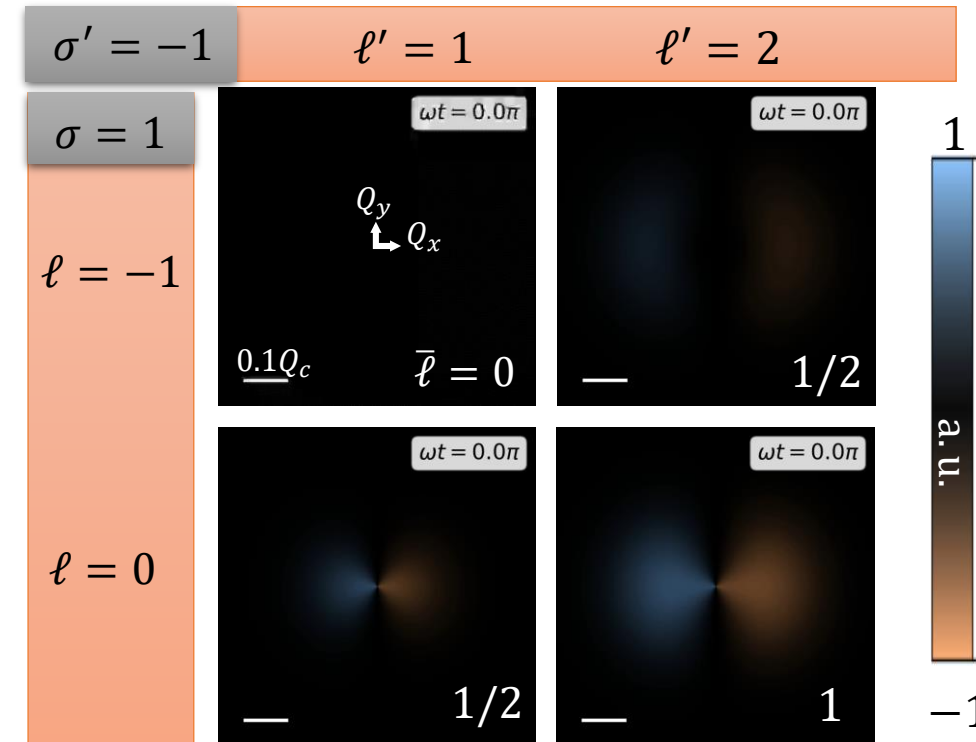
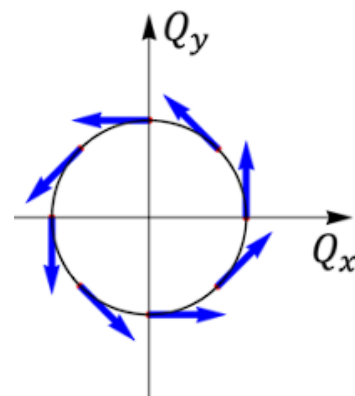
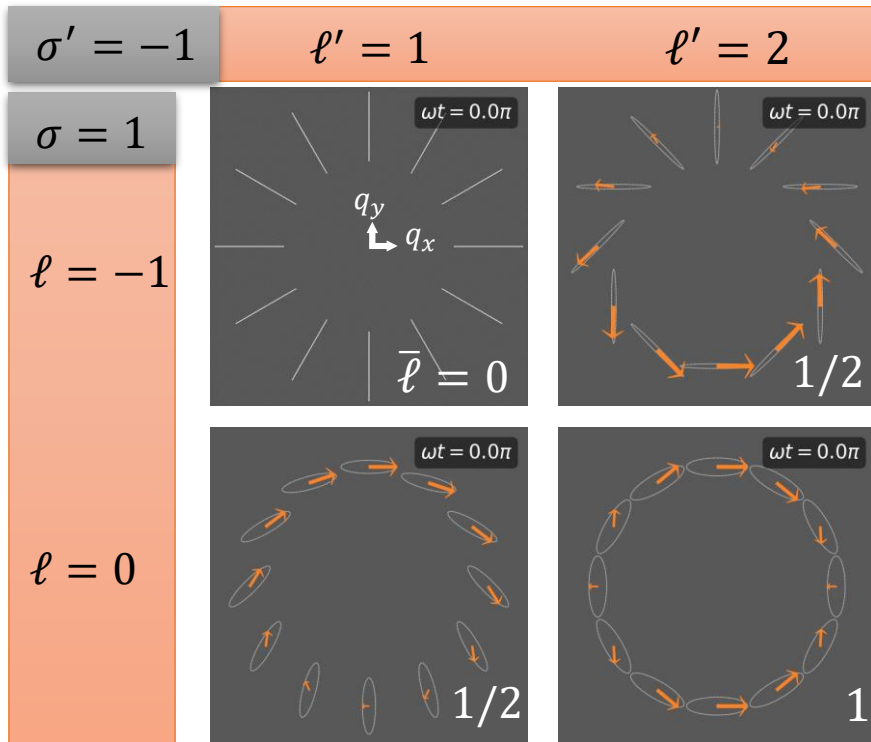
$\bar{\ell} \neq 0$ - rotating ; $\bar{\ell} = 0$ - standing

In the below figures : $\alpha = 0, \beta = \frac{\pi}{2}, \Delta\sigma = -2$

$$\text{Re} \left[\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}, t; \alpha, \beta) \right]$$

$$D^{B-,Q}$$

$$\text{Re} \left[\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{B-,Q}(\mathbf{Q}, t; \alpha, \beta) \right]$$



5. Transition rate of upper bright exciton (beating)

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{B-,Q}(\mathbf{Q}; \alpha, \beta) \propto \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{B-,Q}(\mathbf{Q}; \alpha, \beta) \right|^2 \approx \underbrace{\left| \frac{-iE_g D^B}{4i\hbar} H_{|\ell|,|\ell'|}^+(Q; \beta) \left[\left(1 - \frac{\Delta\sigma}{2}\right) \sin\left(\frac{\Delta\ell - 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - \left(1 + \frac{\Delta\sigma}{2}\right) \sin\left(\frac{\Delta\ell + 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \right|^2}_{\text{node structure Envelope node structure}}$$

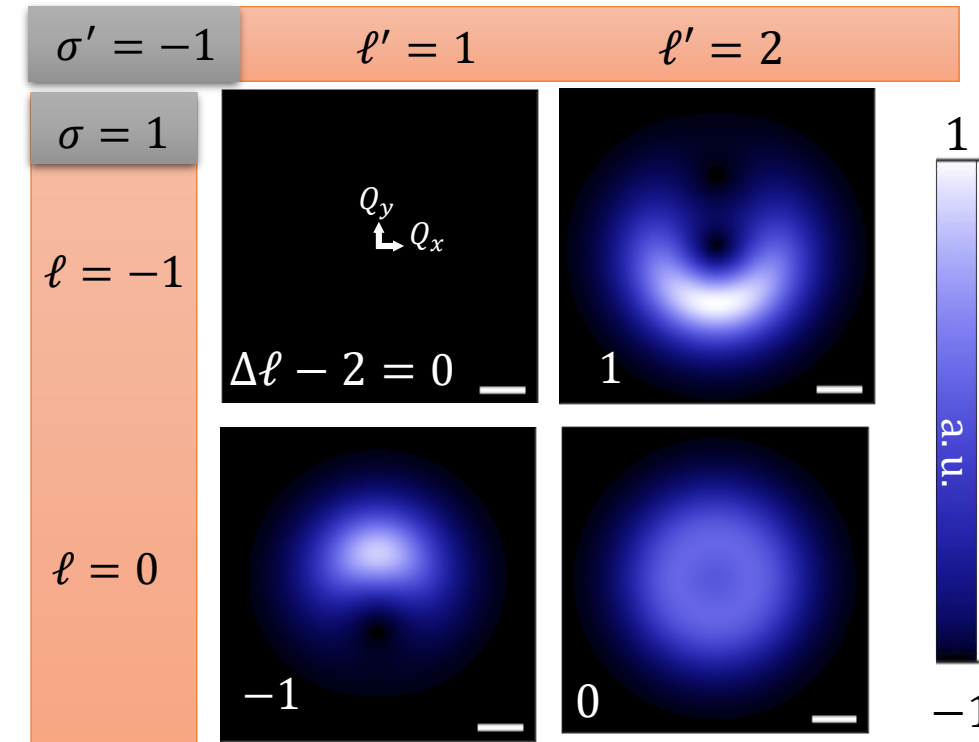
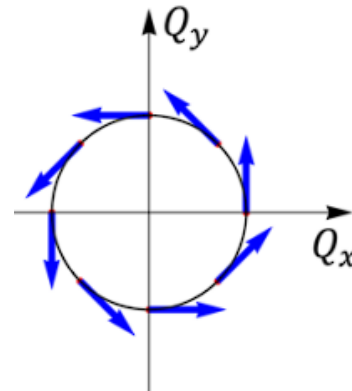
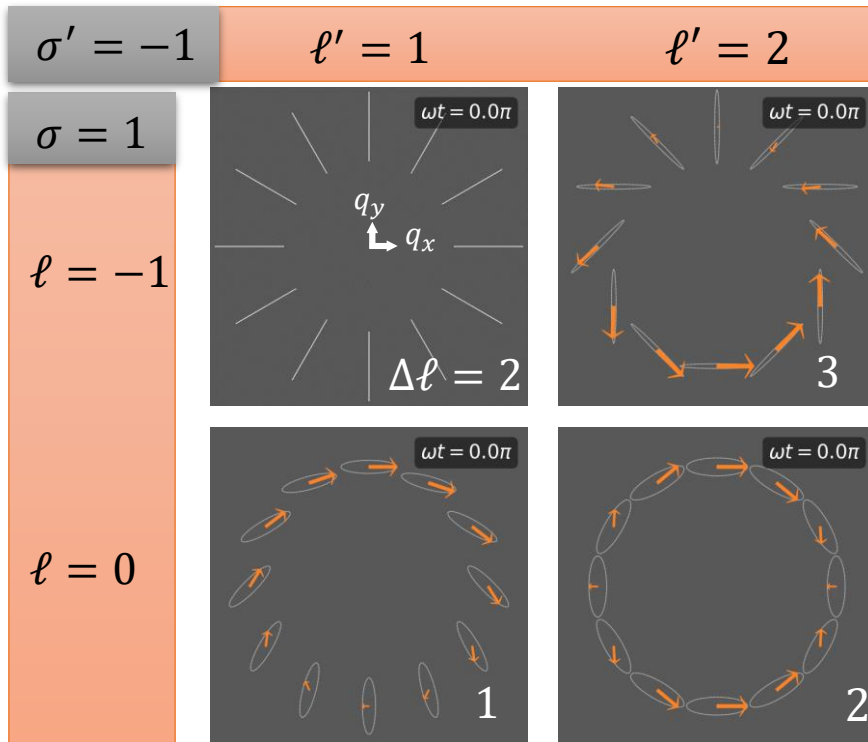
Number of nodes = $\Delta\ell - 2$ (for $\Delta\sigma = -2$)

In the below figures : $\alpha = 0, \beta = \frac{\pi}{2}, \Delta\sigma = -2$

$\text{Re} \left[\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}, t; \alpha, \beta) \right]$

$D^{B-,Q}$

$\Gamma_{\sigma,\ell,\sigma',\ell'}^{B-,Q}(\mathbf{Q}; \alpha, \beta)$

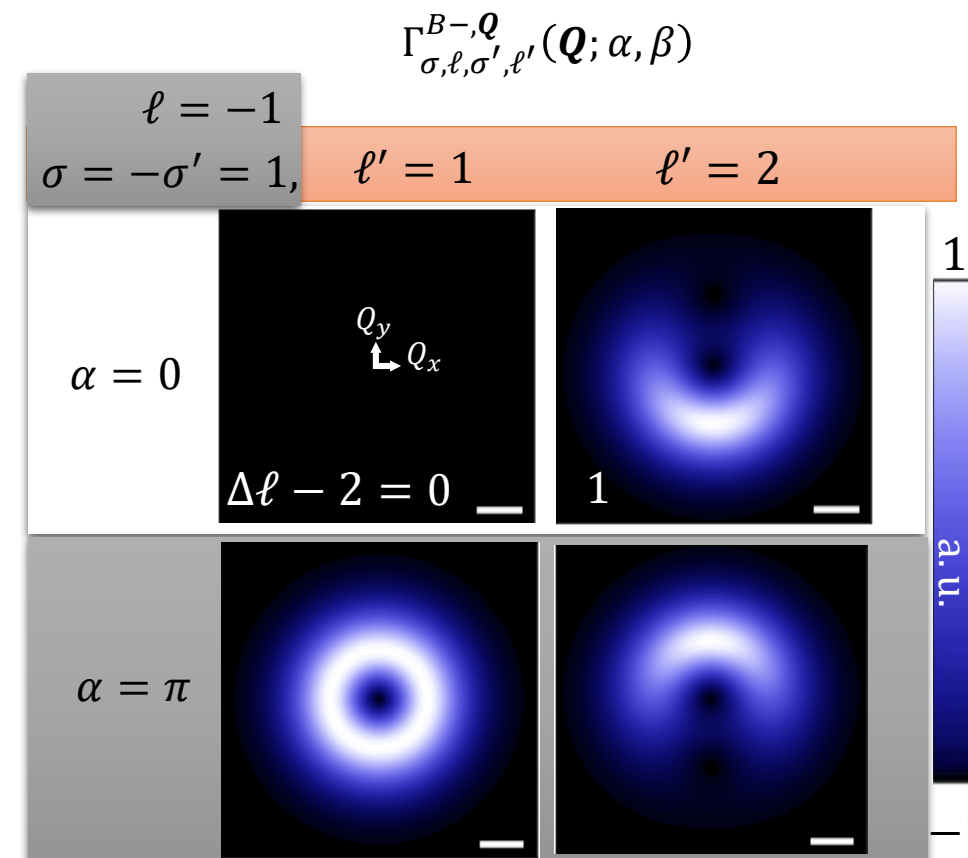
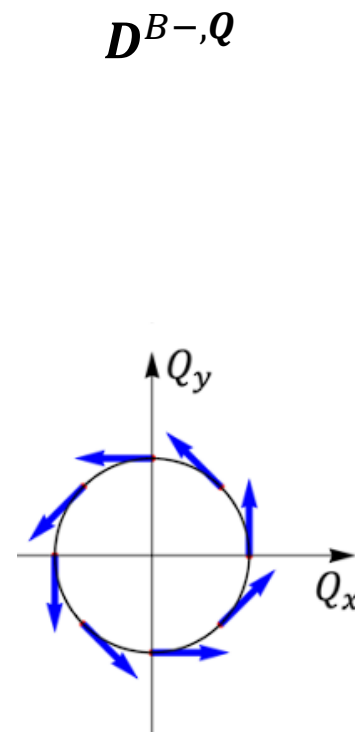
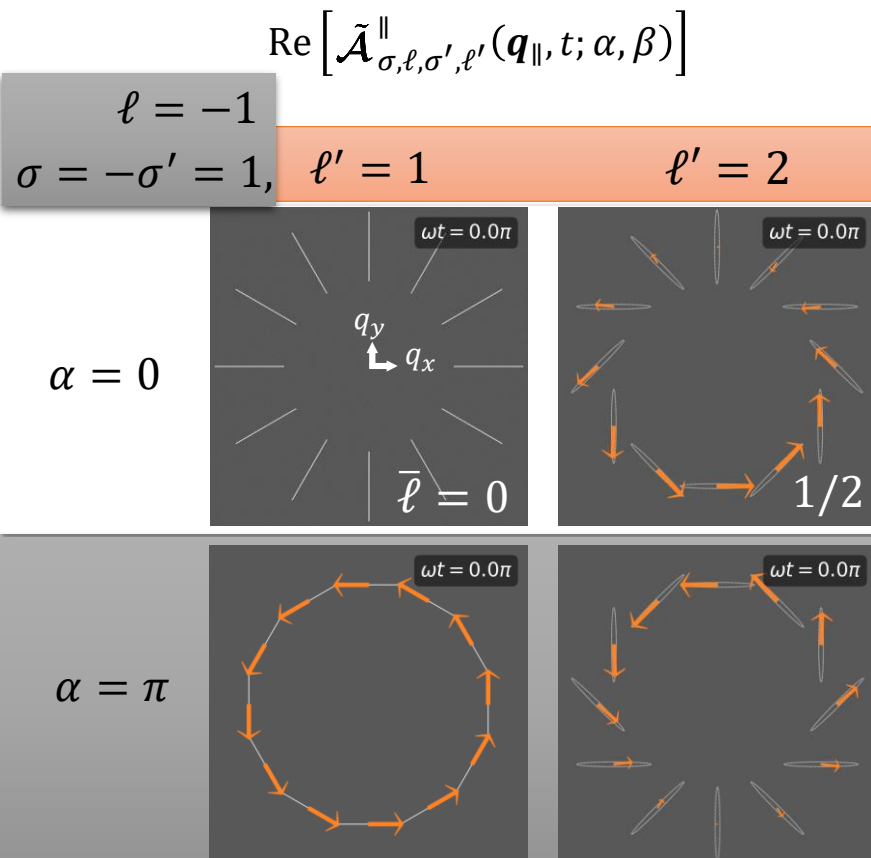


5. Transition rate of upper bright exciton (beating - relative phase α)

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{B-,Q}(\mathbf{Q}; \alpha, \beta) \propto \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{B-,Q}(\mathbf{Q}; \alpha, \beta) \right|^2 \approx \underbrace{\left| \frac{-iE_g D^B}{4i\hbar} H_{|\ell|,|\ell'|}^+(Q; \beta) \left[\left(1 - \frac{\Delta\sigma}{2}\right) \sin\left(\frac{\Delta\ell - 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) - \left(1 + \frac{\Delta\sigma}{2}\right) \sin\left(\frac{\Delta\ell + 2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \right|^2}_{\text{intensity change} \quad \text{Envelope} \quad \text{intensity change}}$$

α change the intensity

In the below figures : $\alpha, \beta = \frac{\pi}{2}, \Delta\sigma = -2$



Total transition rate

6. Total transition rate (B +)

For upper band bright exciton, the total transition rate is obtained as

$$\begin{aligned}
 \Gamma_{\sigma,\ell,\sigma',\ell'}^{B+}(\alpha,\beta) &= \frac{\Omega}{(2\pi)^2} \int_0^\infty \int_0^{2\pi} \Gamma_{\sigma,\ell,\sigma',\ell'}^{B+,Q}(\mathbf{Q}; \alpha, \beta) d\phi_Q Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{4\hbar} \right]^2 \int_0^\infty \int_0^{2\pi} \left\{ \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| + \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[\left(1 - \frac{\Delta\sigma}{2}\right) \cos^2\left(\frac{\Delta\ell-2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \cos^2\left(\frac{\Delta\ell+2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] + \right. \\
 &\quad \left. \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| - \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[\left(1 - \frac{\Delta\sigma}{2}\right) \sin^2\left(\frac{\Delta\ell-2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \sin^2\left(\frac{\Delta\ell+2}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \right\} d\phi_Q Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{4\hbar} \right]^2 \pi \times \int_0^\infty \left\{ \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| + \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[\left(1 - \frac{\Delta\sigma}{2}\right) \left(1 + \delta_{\Delta\ell,2} \cos \alpha_{|\ell|,|\ell'|}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \left(1 + \delta_{\Delta\ell,-2} \cos \alpha_{|\ell|,|\ell'|}\right) \right] + \right. \\
 &\quad \left. \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| - \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[\left(1 - \frac{\Delta\sigma}{2}\right) \left(1 - \delta_{\Delta\ell,2} \cos \alpha_{|\ell|,|\ell'|}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \left(1 - \delta_{\Delta\ell,-2} \cos \alpha_{|\ell|,|\ell'|}\right) \right] \right\} Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{4\hbar} \right]^2 \pi \times 2 \times \\
 &\quad \int_0^\infty \left\{ \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| + \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[1 + \cos \alpha_{|\ell|,|\ell'|} \left(\frac{1}{2} \delta_{\Delta\ell,2} \left(1 - \frac{\Delta\sigma}{2}\right) + \frac{1}{2} \delta_{\Delta\ell,-2} \left(1 + \frac{\Delta\sigma}{2}\right) \right) \right] + \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| - \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[1 - \cos \alpha_{|\ell|,|\ell'|} \left(\frac{1}{2} \delta_{\Delta\ell,2} \left(1 - \frac{\Delta\sigma}{2}\right) + \frac{1}{2} \delta_{\Delta\ell,-2} \left(1 + \frac{\Delta\sigma}{2}\right) \right) \right] \right\} Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{4\hbar} \right]^2 \pi \times 4\pi \left(\frac{A_0 W_0}{\Omega} \right)^2 \times \\
 &\quad \left\{ \left[1 + \sin\beta \frac{\left(\frac{|\ell|+|\ell'|}{2}\right)!}{\sqrt{|\ell|!|\ell'|!}} \right] \left[1 + \cos \alpha_{|\ell|,|\ell'|} \left(\frac{1}{2} \delta_{\Delta\ell,2} \left(1 - \frac{\Delta\sigma}{2}\right) + \frac{1}{2} \delta_{\Delta\ell,-2} \left(1 + \frac{\Delta\sigma}{2}\right) \right) \right] + \left[1 - \sin\beta \frac{\left(\frac{|\ell|+|\ell'|}{2}\right)!}{\sqrt{|\ell|!|\ell'|!}} \right] \left[1 - \cos \alpha_{|\ell|,|\ell'|} \left(\frac{1}{2} \delta_{\Delta\ell,2} \left(1 - \frac{\Delta\sigma}{2}\right) + \frac{1}{2} \delta_{\Delta\ell,-2} \left(1 + \frac{\Delta\sigma}{2}\right) \right) \right] \right\}
 \end{aligned}$$

Rewrite $\left(1 - \frac{\Delta\sigma}{2}\right) = 2\delta_{\Delta\sigma,-2}$, $\left(1 + \frac{\Delta\sigma}{2}\right) = 2\delta_{\Delta\sigma,2}$ and radiant power $\Phi = a_0^2 W_0^2 A_0^2$, $a_0 = \frac{\omega}{\sqrt{2c\mu_0}}$

$$= \frac{\pi\Phi}{4\hbar\Omega} \left[\frac{E_g D_B^X}{a_0 \hbar} \right]^2 \times \left[1 + (\delta_{\Delta\ell,2} \delta_{\Delta\sigma,-2} + \delta_{\Delta\ell,-2} \delta_{\Delta\sigma,2}) \cos \alpha_{|\ell|,|\ell'|} \sin\beta \frac{\left(\frac{|\ell|+|\ell'|}{2}\right)!}{\sqrt{|\ell|!|\ell'|!}} \right]$$

Winding of light  **Winding number** **spin**

When polarization (spin and winding of light) matches transition dipole (winding of exciton), α switches the transition across the entire ϕ_Q angle, thus tuning the total transition rate.

6. Total transition rate (B —)

For lower band bright exciton, the total transition rate is obtained as

$$\begin{aligned}
 \Gamma_{\sigma,\ell,\sigma',\ell'}^{\mathbf{B}-}(\alpha,\beta) &= \frac{\Omega}{(2\pi)^2} \int_0^\infty \int_0^{2\pi} \Gamma_{\sigma,\ell,\sigma',\ell'}^{\mathbf{B}-,\mathbf{Q}}(\mathbf{Q};\alpha,\beta) d\phi_{\mathbf{Q}} Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{4\hbar} \right]^2 \int_0^\infty \int_0^{2\pi} \left\{ \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| + \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[\left(1 - \frac{\Delta\sigma}{2}\right) \sin^2\left(\frac{\Delta\ell-2}{2}\phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \sin^2\left(\frac{\Delta\ell+2}{2}\phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] + \right. \\
 &\quad \left. \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| - \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[\left(1 - \frac{\Delta\sigma}{2}\right) \cos^2\left(\frac{\Delta\ell-2}{2}\phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \cos^2\left(\frac{\Delta\ell+2}{2}\phi_{\mathbf{Q}} + \frac{\alpha_{|\ell|,|\ell'|}}{2}\right) \right] \right\} d\phi_{\mathbf{Q}} Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{4\hbar} \right]^2 \pi \times \int_0^\infty \left\{ \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| + \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[\left(1 - \frac{\Delta\sigma}{2}\right) \left(1 - \delta_{\Delta\ell,2} \cos \alpha_{|\ell|,|\ell'|}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \left(1 - \delta_{\Delta\ell,-2} \cos \alpha_{|\ell|,|\ell'|}\right) \right] + \right. \\
 &\quad \left. \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| - \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[\left(1 - \frac{\Delta\sigma}{2}\right) \left(1 + \delta_{\Delta\ell,2} \cos \alpha_{|\ell|,|\ell'|}\right) + \left(1 + \frac{\Delta\sigma}{2}\right) \left(1 + \delta_{\Delta\ell,-2} \cos \alpha_{|\ell|,|\ell'|}\right) \right] \right\} Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{4\hbar} \right]^2 \pi \times 2 \times \\
 &\quad \int_0^\infty \left\{ \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| + \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[1 - \cos \alpha_{|\ell|,|\ell'|} \left(\frac{1}{2} \delta_{\Delta\ell,2} \left(1 - \frac{\Delta\sigma}{2}\right) + \frac{1}{2} \delta_{\Delta\ell,-2} \left(1 + \frac{\Delta\sigma}{2}\right) \right) \right] + \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| - \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left[1 + \cos \alpha_{|\ell|,|\ell'|} \left(\frac{1}{2} \delta_{\Delta\ell,2} \left(1 - \frac{\Delta\sigma}{2}\right) + \frac{1}{2} \delta_{\Delta\ell,-2} \left(1 + \frac{\Delta\sigma}{2}\right) \right) \right] \right\} Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{4\hbar} \right]^2 \pi \times 4\pi \left(\frac{A_0 W_0}{\Omega} \right)^2 \times \\
 &\quad \left\{ \left[1 + \sin\beta \frac{\left(\frac{|\ell|+|\ell'|}{2}\right)!}{\sqrt{|\ell|!|\ell'|!}} \right] \left[1 - \cos \alpha_{|\ell|,|\ell'|} \left(\frac{1}{2} \delta_{\Delta\ell,2} \left(1 - \frac{\Delta\sigma}{2}\right) + \frac{1}{2} \delta_{\Delta\ell,-2} \left(1 + \frac{\Delta\sigma}{2}\right) \right) \right] + \left[1 - \sin\beta \frac{\left(\frac{|\ell|+|\ell'|}{2}\right)!}{\sqrt{|\ell|!|\ell'|!}} \right] \left[1 + \cos \alpha_{|\ell|,|\ell'|} \left(\frac{1}{2} \delta_{\Delta\ell,2} \left(1 - \frac{\Delta\sigma}{2}\right) + \frac{1}{2} \delta_{\Delta\ell,-2} \left(1 + \frac{\Delta\sigma}{2}\right) \right) \right] \right\}
 \end{aligned}$$

Rewrite $\left(1 - \frac{\Delta\sigma}{2}\right) = 2\delta_{\Delta\sigma,-2}$, $\left(1 + \frac{\Delta\sigma}{2}\right) = 2\delta_{\Delta\sigma,2}$ and radiant power $\Phi = a_0^2 W_0^2 A_0^2$, $a_0 = \frac{\omega}{\sqrt{2c\mu_0}}$

$$= \frac{\pi\Phi}{4\hbar\Omega} \left[\frac{E_g D_B^X}{a_0\hbar} \right]^2 \times \left[1 - (\delta_{\Delta\ell,2} \delta_{\Delta\sigma,-2} + \delta_{\Delta\ell,-2} \delta_{\Delta\sigma,2}) \cos \alpha_{|\ell|,|\ell'|} \sin\beta \frac{\left(\frac{|\ell|+|\ell'|}{2}\right)!}{\sqrt{|\ell|!|\ell'|!}} \right]$$

Winding of light
 spin
 Winding number

When polarization (spin and winding of light) matches transition dipole (winding of exciton), α switches the transition across the entire $\phi_{\mathbf{Q}}$ angle, thus tuning the total transition rate.

6. Total transition rate (G)

For gray exciton, the total transition rate is obtained as

$$\begin{aligned}
 \Gamma_{\sigma,\ell,\sigma',\ell'}^G(\alpha, \beta) &= \frac{\Omega}{(2\pi)^2} \int_0^\infty \int_0^{2\pi} \Gamma_{\sigma,\ell,\sigma',\ell'}^{G,Q}(\mathbf{Q}; \alpha, \beta) d\phi_Q Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{2\sqrt{2}\hbar} \right]^2 \int_0^\infty \int_0^{2\pi} \sin^2 \theta_q \left\{ \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| + \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \cos^2 \left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2} \right) + \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| - \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \sin^2 \left(\frac{\Delta J}{2} \phi_Q + \frac{\alpha_{|\ell|,|\ell'|}}{2} \right) \right\} d\phi_Q Q dQ \\
 \sin \theta_q &\approx \frac{Q}{q_0} \\
 &\approx \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{2\sqrt{2}\hbar} \right]^2 \times \pi \int_0^\infty \left\{ \left(\frac{Q}{q_0} \right)^2 \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| + \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left(1 + \delta_{\Delta J,0} \cos \alpha_{|\ell|,|\ell'|} \right) + \left(\frac{Q}{q_0} \right)^2 \left[\left| \cos\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell|}(Q) \right| - \left| \sin\left(\frac{\beta}{2}\right) \tilde{F}_{|\ell'|}(Q) \right| \right]^2 \left(1 - \delta_{\Delta J,0} \cos \alpha_{|\ell|,|\ell'|} \right) \right\} Q dQ \\
 &= \frac{\Omega}{4\pi^2} \frac{2\pi}{\hbar} \left[\frac{E_g D_B^X}{2\sqrt{2}\hbar} \right]^2 \times \pi \times 4\pi \left(\frac{A_0}{\Omega q_0} \right)^2 \times \\
 &\quad \left\{ \left[1 + \cos^2\left(\frac{\beta}{2}\right) |\ell| + \sin^2\left(\frac{\beta}{2}\right) |\ell'| + \sin\beta \frac{\left(\frac{|\ell|+|\ell'|}{2}+1\right)!}{\sqrt{|\ell|!|\ell'|!}} \right] \left(1 + \delta_{\Delta J,0} \cos \alpha_{|\ell|,|\ell'|} \right) + \left[1 + \cos^2\left(\frac{\beta}{2}\right) |\ell| + \sin^2\left(\frac{\beta}{2}\right) |\ell'| - \sin\beta \frac{\left(\frac{|\ell|+|\ell'|}{2}+1\right)!}{\sqrt{|\ell|!|\ell'|!}} \right] \left(1 - \delta_{\Delta J,0} \cos \alpha_{|\ell|,|\ell'|} \right) \right\}
 \end{aligned}$$

$$\text{radiant power } \Phi = a_0^2 W_0^2 A_0^2, \quad a_0 = \frac{\omega}{\sqrt{2c\mu_0}}$$

$$= \frac{\pi\Phi}{2\hbar\Omega} \left[\frac{E_g D_B^X}{a_0 q_0 W_0 \hbar} \right]^2 \times \left\{ \left[1 + \cos^2\left(\frac{\beta}{2}\right) |\ell| + \sin^2\left(\frac{\beta}{2}\right) |\ell'| \right] + \delta_{\Delta J,0} \cos \alpha_{|\ell|,|\ell'|} \sin\beta \frac{\left(\frac{|\ell|+|\ell'|}{2}+1\right)!}{\sqrt{|\ell|!|\ell'|!}} \right\}$$

↓
Longitudinal component
increase it's intensity as
 $|\ell|$ or $|\ell'|$ increase

↓
Beating

When coherent, $\Delta J = 0$, α controls constructive
and destructive interference

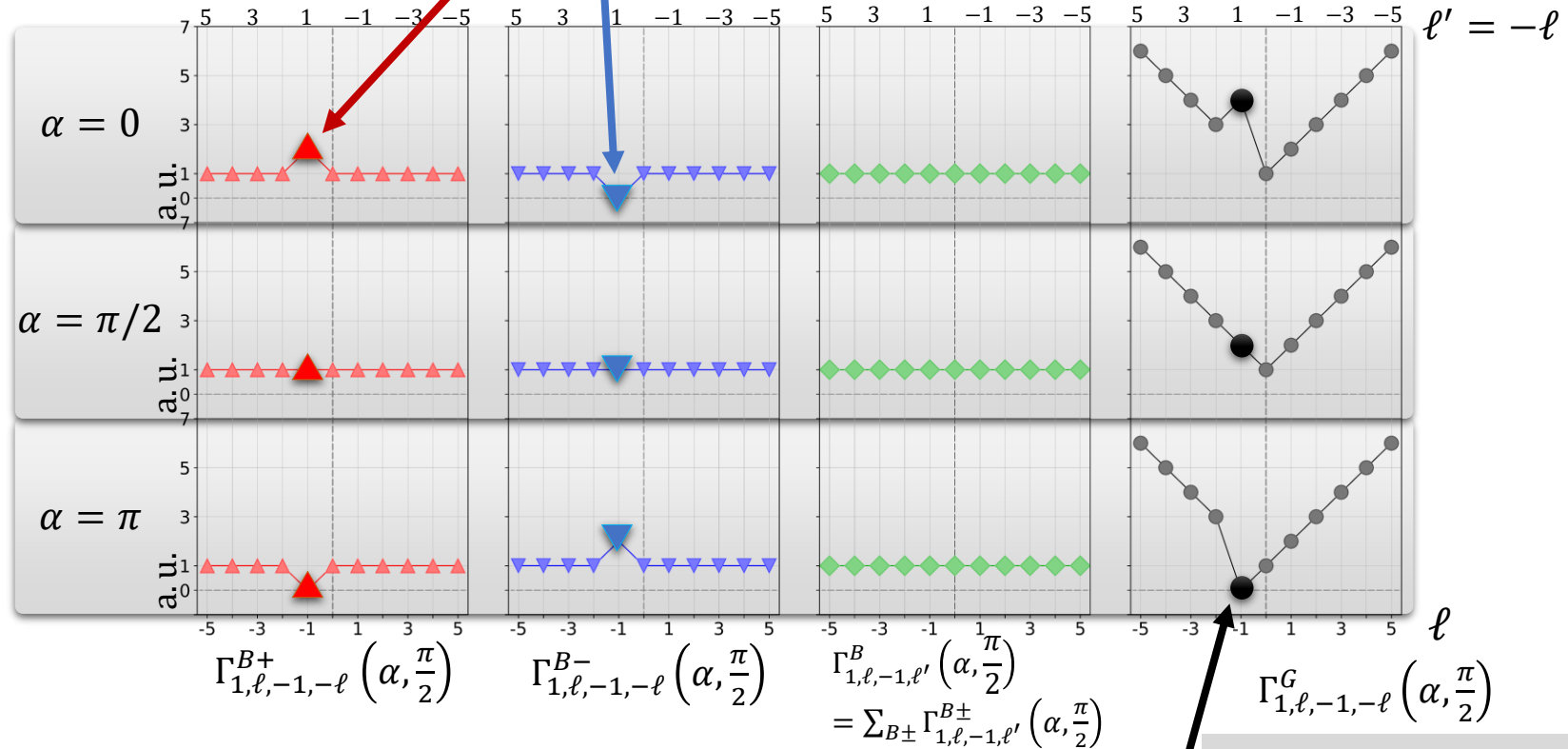
radiant power
 $\Phi = a_0^2 W_0^2 A_0^2,$
 $a_0 = \frac{\omega}{\sqrt{2c\mu_0}}$

6. Total Transition Rate

For $B \pm$: $\Gamma_{\sigma,\ell,\sigma',\ell'}^{B\pm}(\alpha,\beta) = \frac{\pi\Phi}{4\hbar\Omega} \left[\frac{E_g D_B^X}{a_0 \hbar} \right]^2 \times \left[1 \pm (\delta_{\Delta\ell,2} \delta_{\Delta\sigma,-2} + \delta_{\Delta\ell,-2} \delta_{\Delta\sigma,2}) \cos \alpha_{|\ell|,|\ell'|} \sin \beta \frac{\left(\frac{|\ell|+|\ell'|}{2} \right)!}{\sqrt{|\ell|!|\ell'|!}} \right]$

Condition for α control :

Spin ($\Delta\sigma$) and winding of light ($\Delta\ell$) match winding of exciton (2)



Condition for α control :

Coherent light beams, $\Delta J = 0$

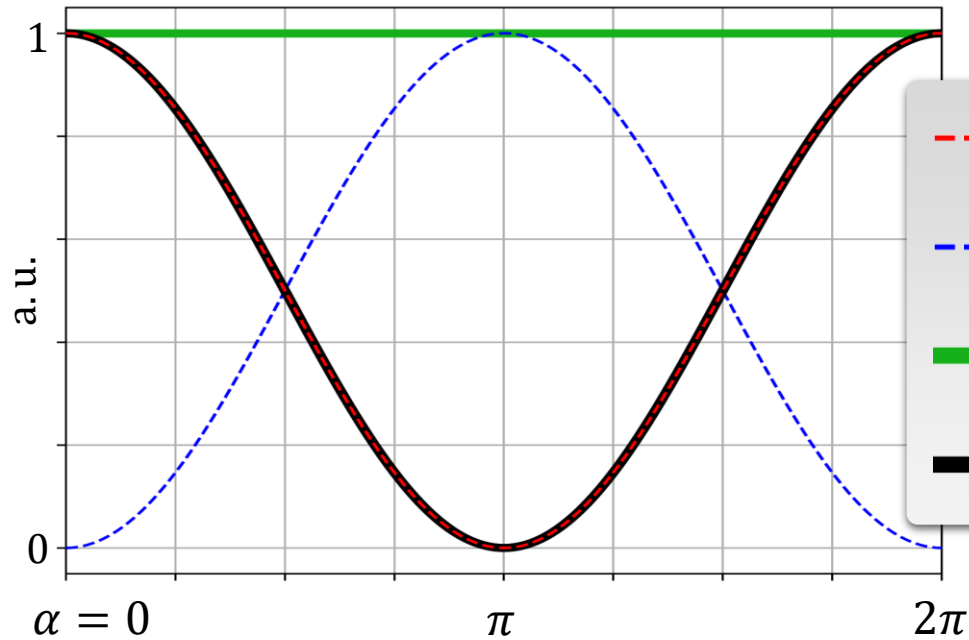
For $G \pm$: $\Gamma_{\sigma,\ell,\sigma',\ell'}^G(\alpha,\beta) = \frac{\pi\Phi}{2\hbar\Omega} \left[\frac{E_g D_B^X}{a_0 q_0 W_0 \hbar} \right]^2 \times \left\{ \left[1 + \cos^2 \left(\frac{\beta}{2} \right) |\ell| + \sin^2 \left(\frac{\beta}{2} \right) |\ell'| \right] + \delta_{\Delta J,0} \cos \alpha_{|\ell|,|\ell'|} \sin \beta \frac{\left(\frac{|\ell|+|\ell'|}{2} + 1 \right)!}{\sqrt{|\ell|!|\ell'|!}} \right\}$

6. Total Transition Rate ($\sigma, \ell, \sigma', \ell' = 1, -1, -1, 1$)

For $B \pm$: $\Gamma_{\sigma, \ell, \sigma', \ell'}^{B\pm}(\alpha, \beta) = \frac{\pi\Phi}{4\hbar\Omega} \left[\frac{E_g D_B^X}{a_0 \hbar} \right]^2 \times \left[1 \pm (\delta_{\Delta\ell, 2} \delta_{\Delta\sigma, -2} + \delta_{\Delta\ell, -2} \delta_{\Delta\sigma, 2}) \cos \alpha_{|\ell|, |\ell'|} \sin \beta \frac{\left(\frac{|\ell| + |\ell'|}{2} \right)!}{\sqrt{|\ell|! |\ell'|!}} \right]$

For $G \pm$: $\Gamma_{\sigma, \ell, \sigma', \ell'}^G(\alpha, \beta) = \frac{\pi\Phi}{2\hbar\Omega} \left[\frac{E_g D_B^X}{a_0 q_0 W_0 \hbar} \right]^2 \times \left\{ \left[1 + \cos^2 \left(\frac{\beta}{2} \right) |\ell| + \sin^2 \left(\frac{\beta}{2} \right) |\ell'| \right] + \delta_{\Delta J, 0} \cos \alpha_{|\ell|, |\ell'|} \sin \beta \frac{\left(\frac{|\ell| + |\ell'|}{2} + 1 \right)!}{\sqrt{|\ell|! |\ell'|!}} \right\}$

$\Delta\ell = 2$ (winding number),
 $\Delta\sigma = -2$,
and $\Delta J = 0$



$\Gamma_{1,-1,-1,1}^{B+} \left(\alpha, \frac{\pi}{2} \right)$
 $\Gamma_{1,-1,-1,1}^{B-} \left(\alpha, \frac{\pi}{2} \right)$
 $\Gamma_{1,-1,-1,1}^B \left(\alpha, \frac{\pi}{2} \right)$
 $\Gamma_{1,-1,-1,1}^G \left(\alpha, \frac{\pi}{2} \right)$

When $\sigma, \ell, \sigma', \ell' = 1, -1, -1, 1$,
the transition rate is independent of ϕ_Q ; thus,
 α change the total transition

$\Gamma_{1,-1,-1,1}^{B+} \left(\alpha, \frac{\pi}{2} \right)$
 $\Gamma_{1,-1,-1,1}^{B-} \left(\alpha, \frac{\pi}{2} \right)$
 $\Gamma_{1,-1,-1,1}^G \left(\alpha, \frac{\pi}{2} \right)$

